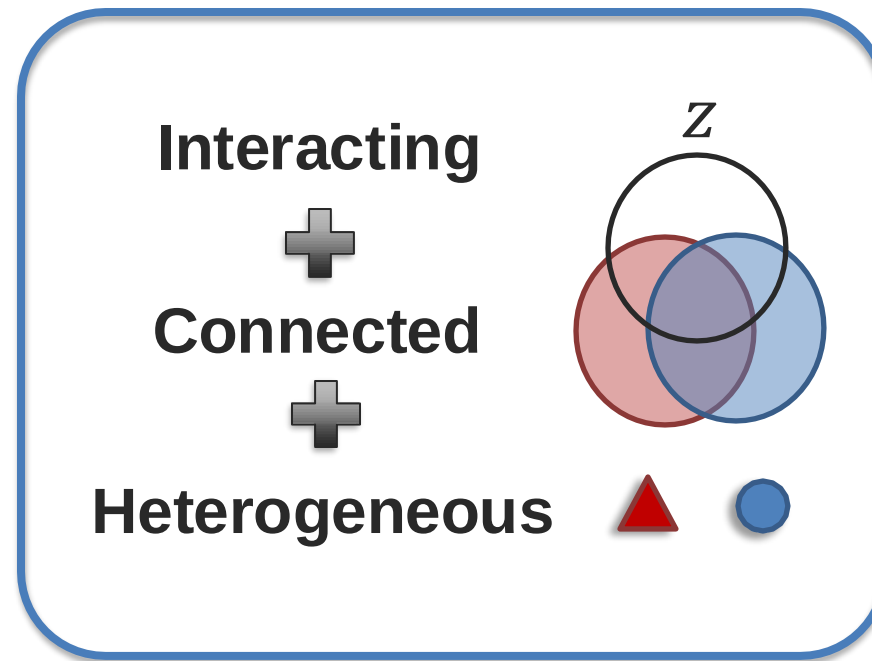




Lecture 5: Multimodal Representation (Part 3)

Email: idanschwartz@gmail.com or
idan.schwartz@biu.ac.il

Multimodal is the scientific study of **heterogeneous**,
connected and **interacting** data

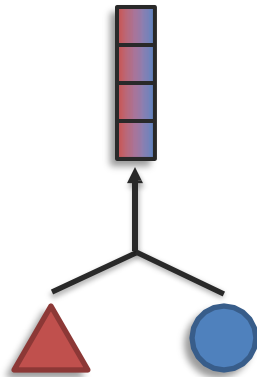


How to model interactions?

Multimodal Representation

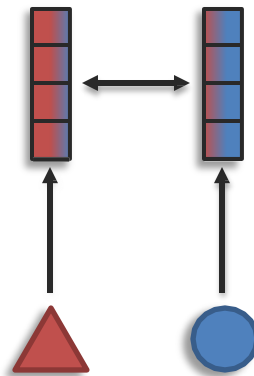
Definition: Learning representations that reflect cross-modal interactions between individual elements, across different modalities

Fusion



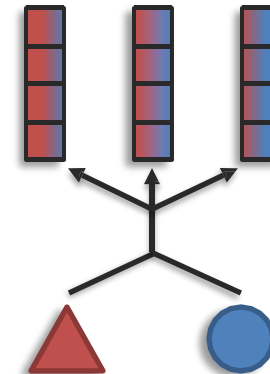
modalities $>$ # representations

Coordination



modalities $=$ # representations

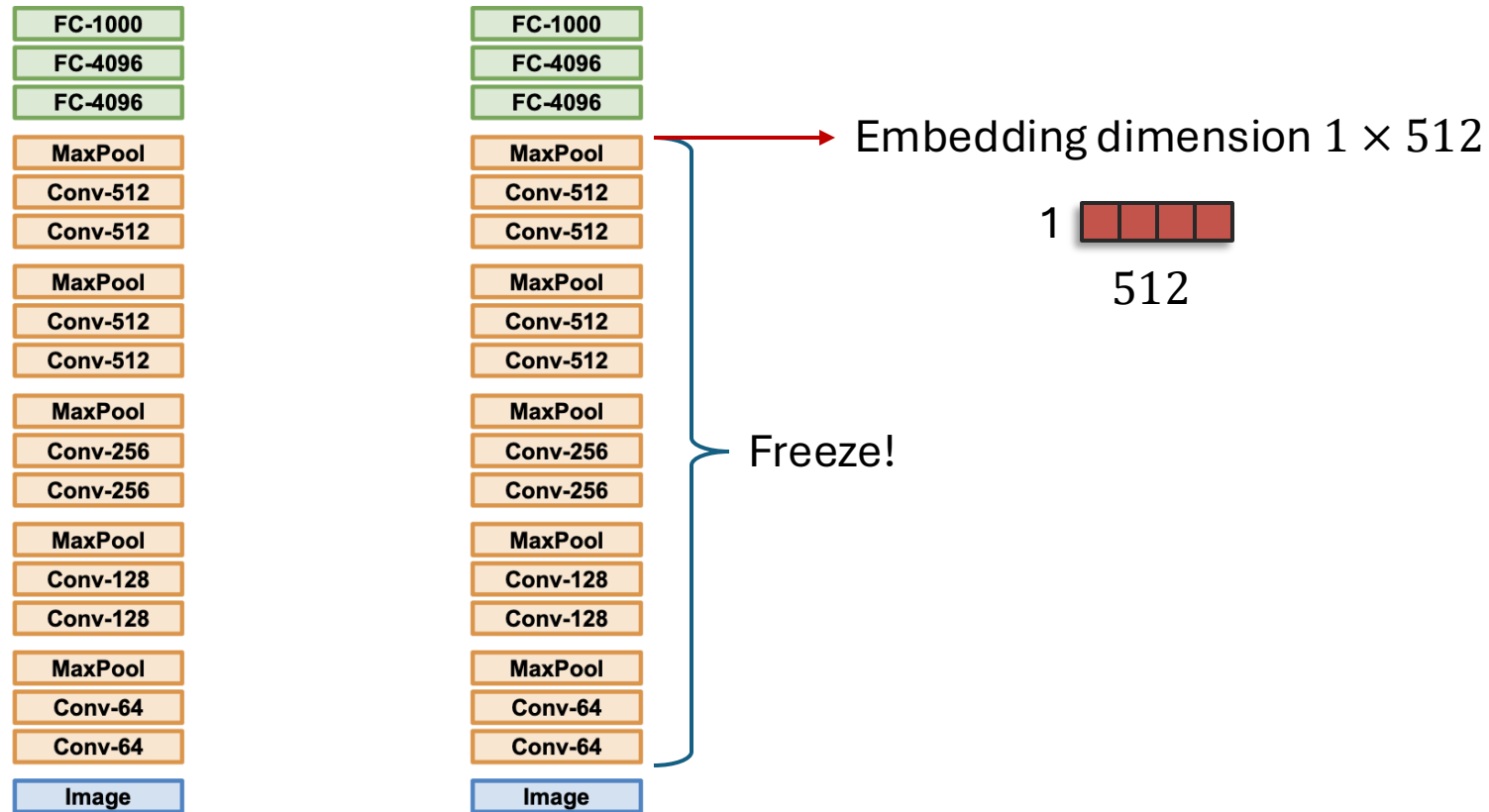
Factorization



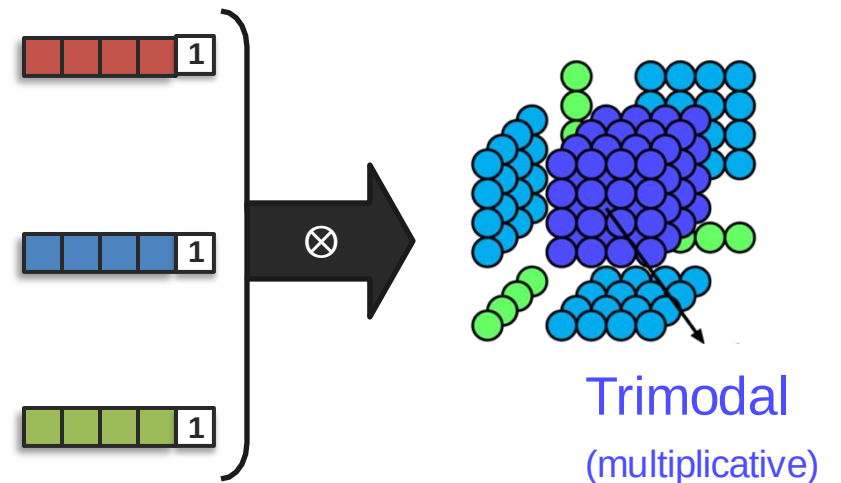
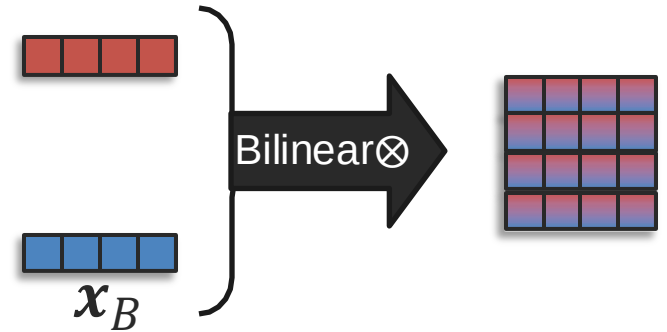
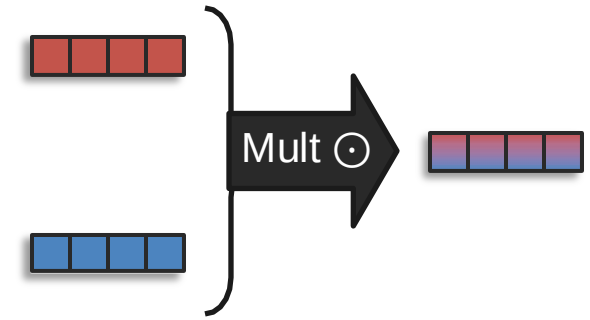
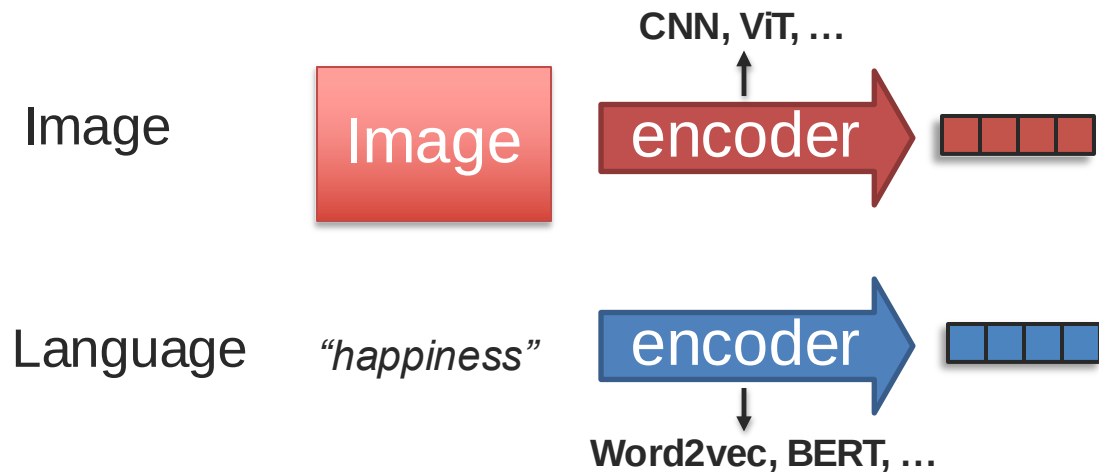
modalities $<$ # representations

Transfer Learning

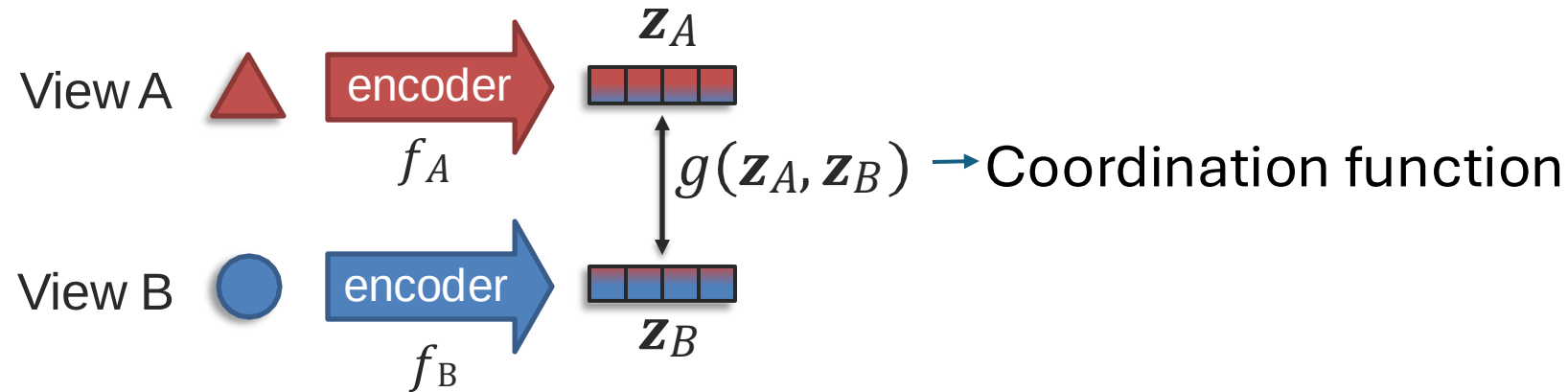
1. Train on ImageNet 2. Extract features



Fuse representations



Coordinate representations



E.g., cosine similarity:

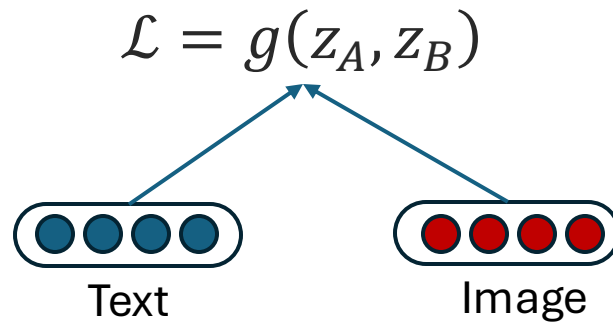
$$g(z_A, z_B) = \frac{z_A \cdot z_B}{\|z_A\| \|z_B\|}$$

Learning with coordination function:

$$\mathcal{L} = g(z_A, z_B)$$

Learning with coordination function

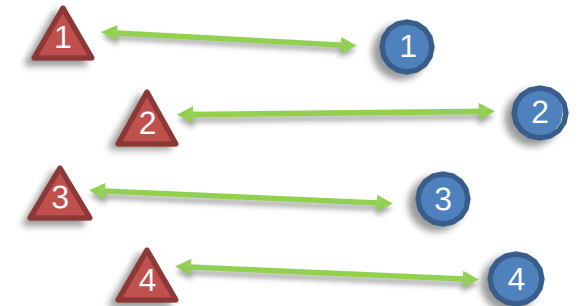
- Neural Network:



- Optimize w/ gradient descent: $\min_{Z_A, Z_B} \mathcal{L}$

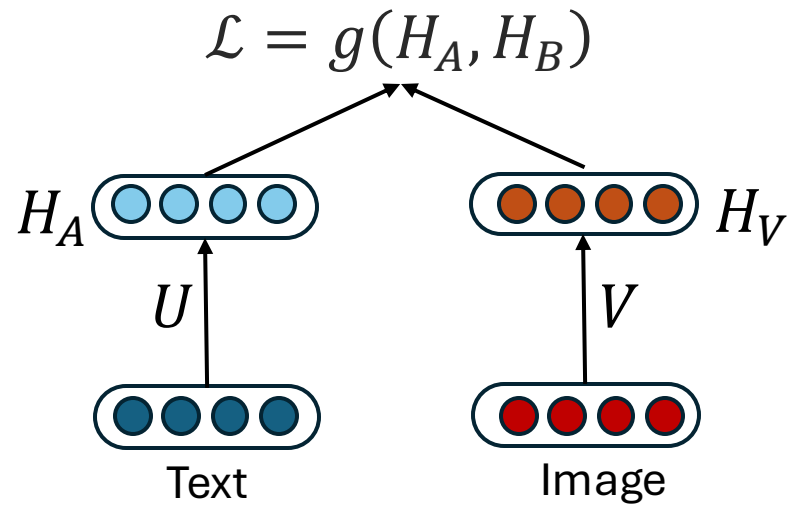
Paired data: {, }

(e.g., images and text descriptions)



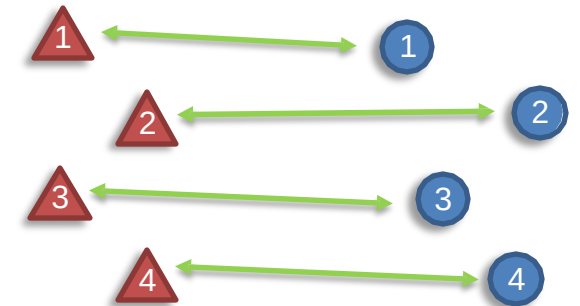
CCA

- Neural Network:



Paired data: {, }

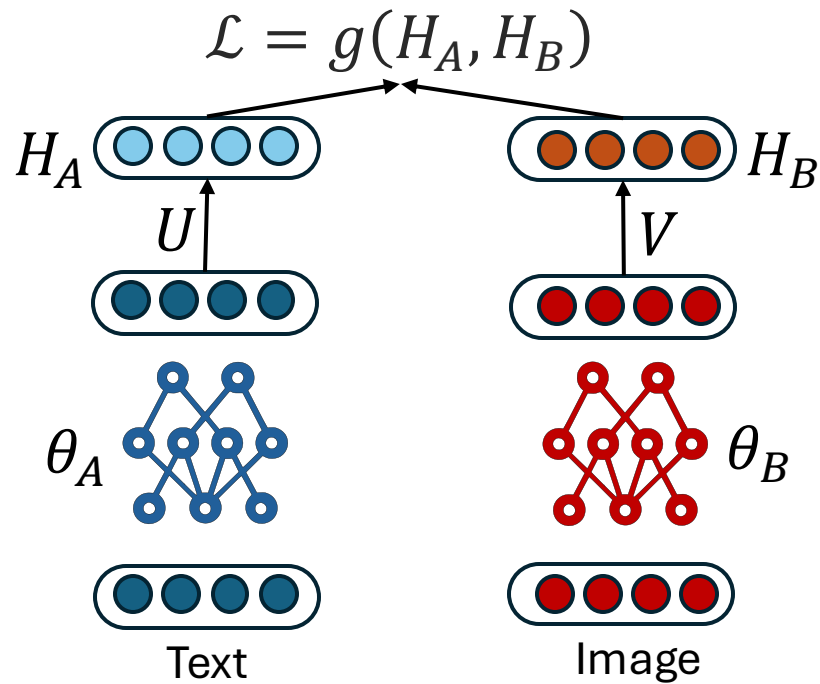
(e.g., images and text descriptions)



- Optimize w/ gradient descent: $\min_{U,V} \mathcal{L}$

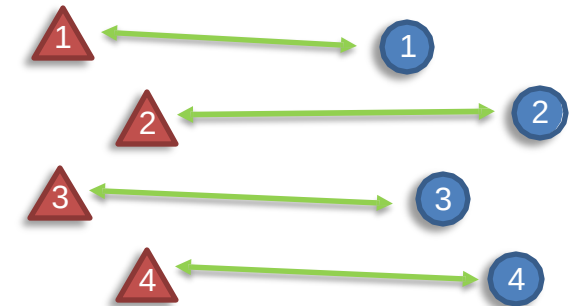
DCCA

- Neural Network:



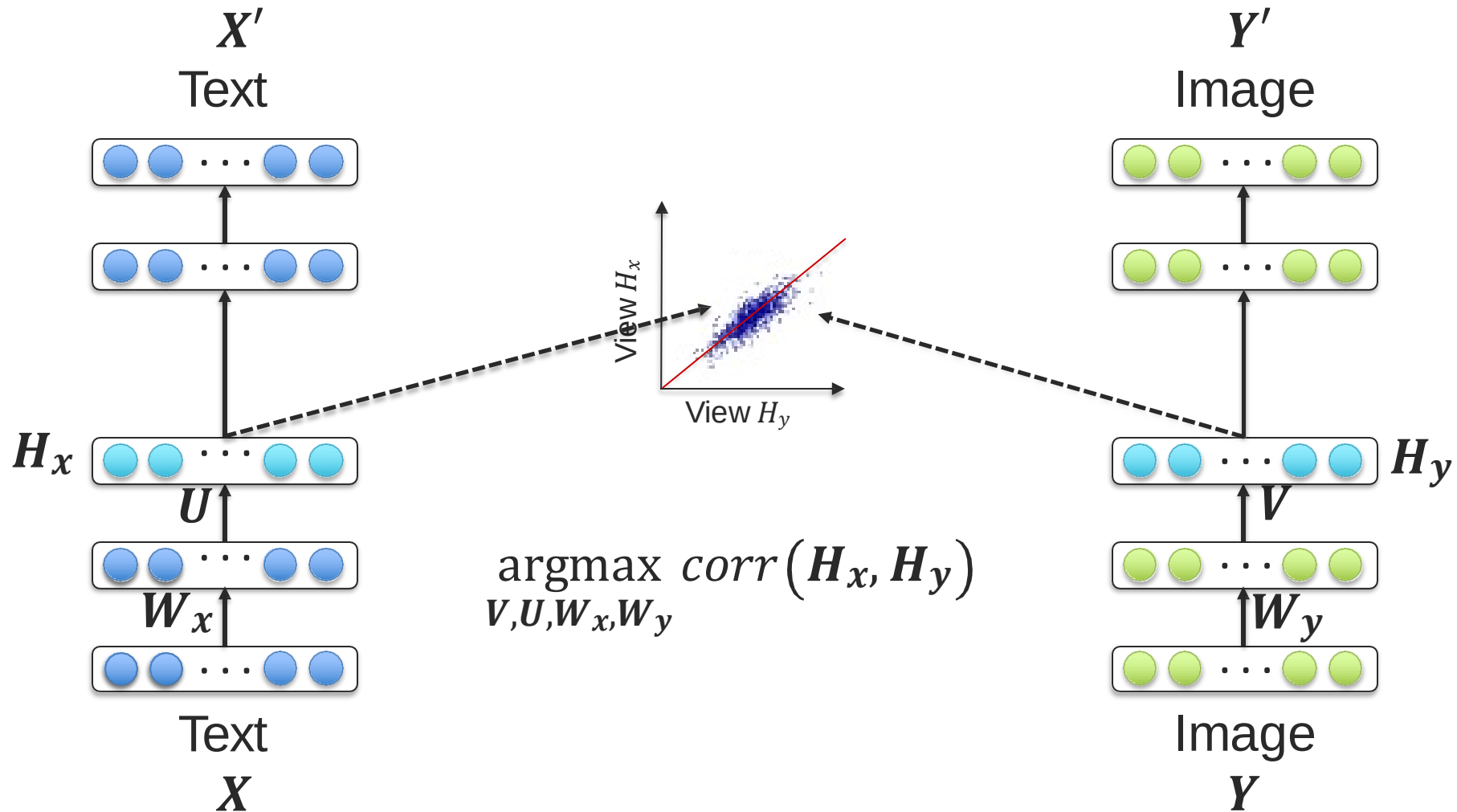
Paired data: {, }

(e.g., images and text descriptions)

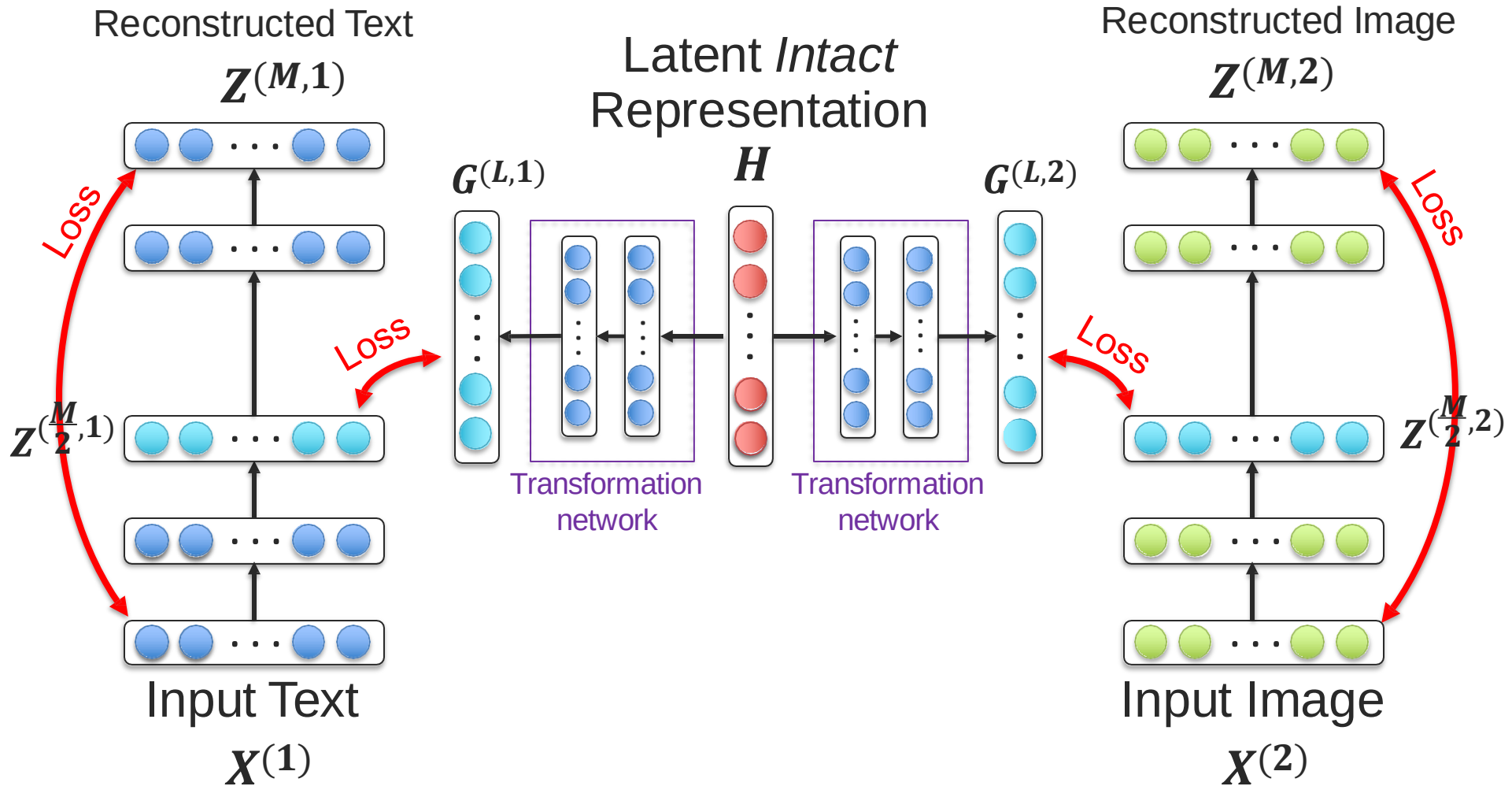


- Optimize w/ gradient descent: $\min_{\theta_A, \theta_B, U, V} \mathcal{L}$

Can be part of a larger architecture



And employ an advanced coordination function



Contrastive loss

- Contrastive loss:

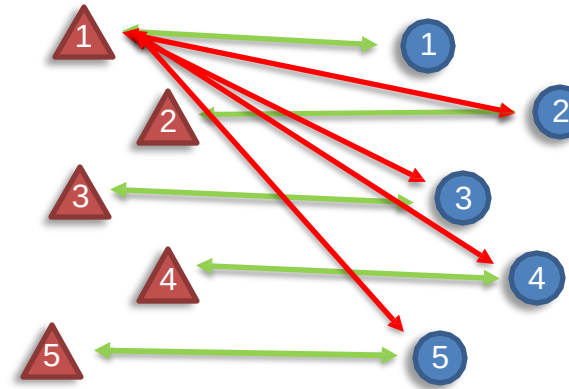
- brings **positive pairs** closer
- pushes **negative pairs** apart

- Simple contrastive loss:

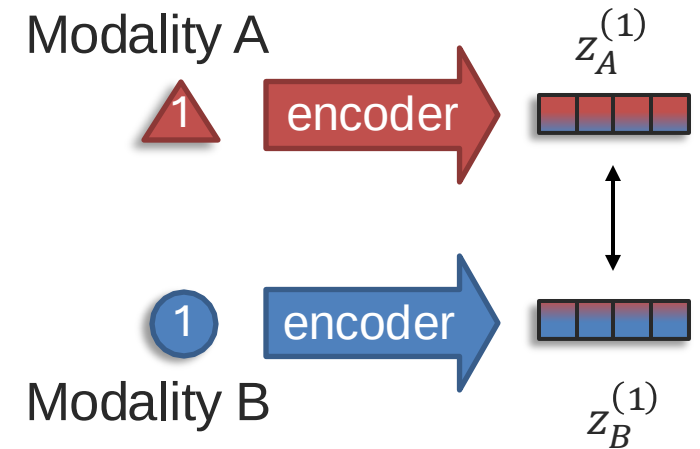
$$\mathcal{L} = \max \left\{ \alpha + \underline{\text{sim} \left(z_A^{(i)}, z_B^{(i)} \right)} - \underline{\text{sim} \left(z_A^{(i)}, z_B^{(j)} \right)}, 0 \right\}$$

- Similarity is often measured with cosine operator

Multimodal data: { ,  }



 Positive pairs
 Negative pairs

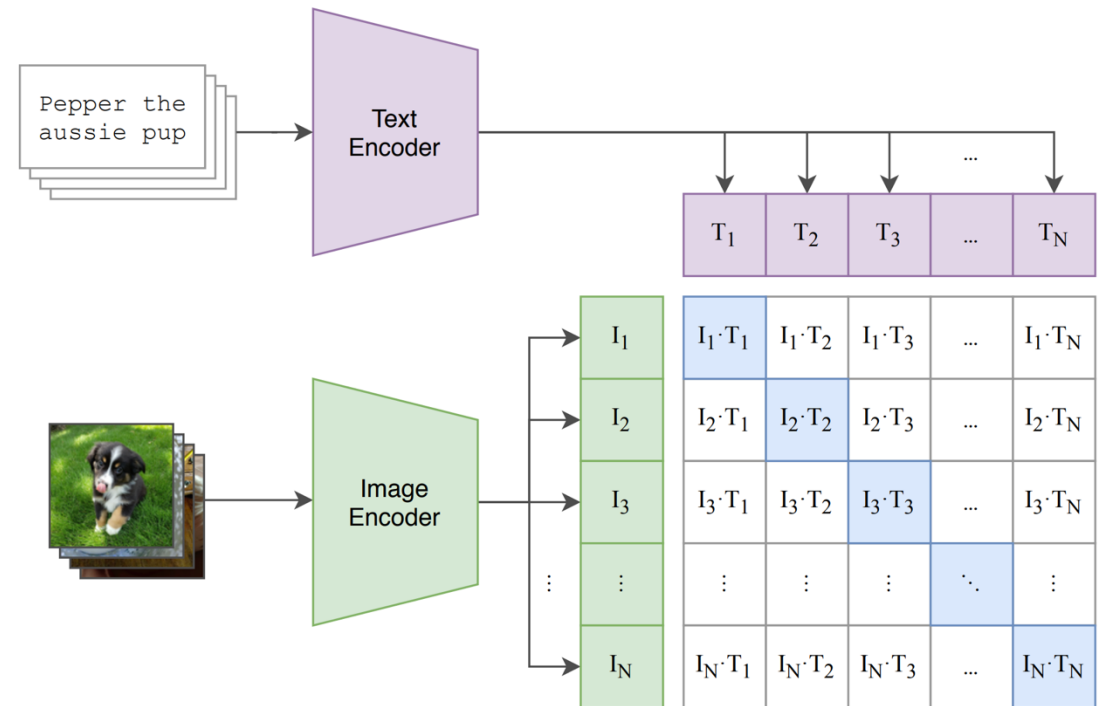
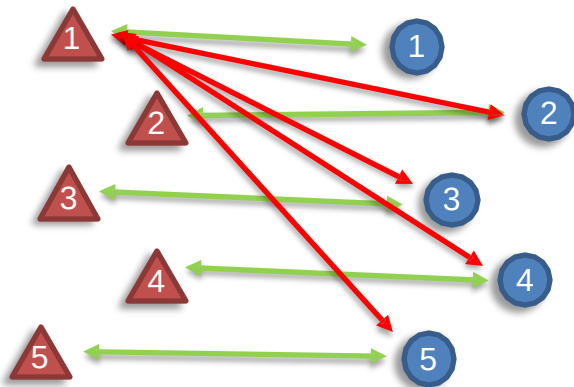


Representations based on Contrastive Language–Image Pre-training (CLIP)

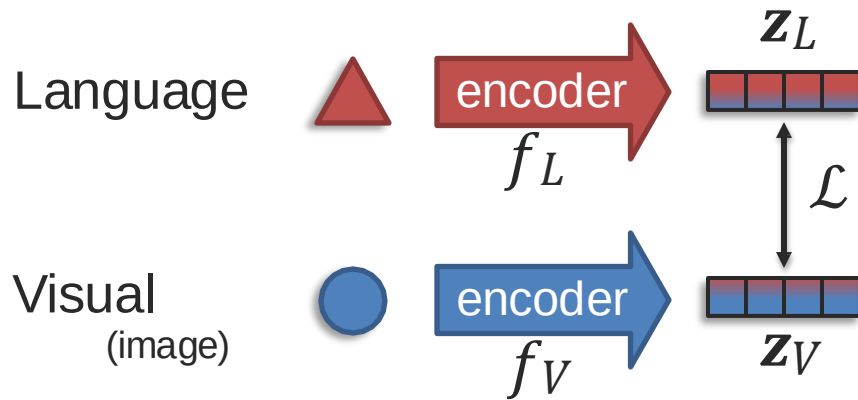
- Dataset of **400 million (image, text)** pairs collected from the internet

Contrastive loss:

- brings **positive pairs** closer
- pushes **negative pairs** apart

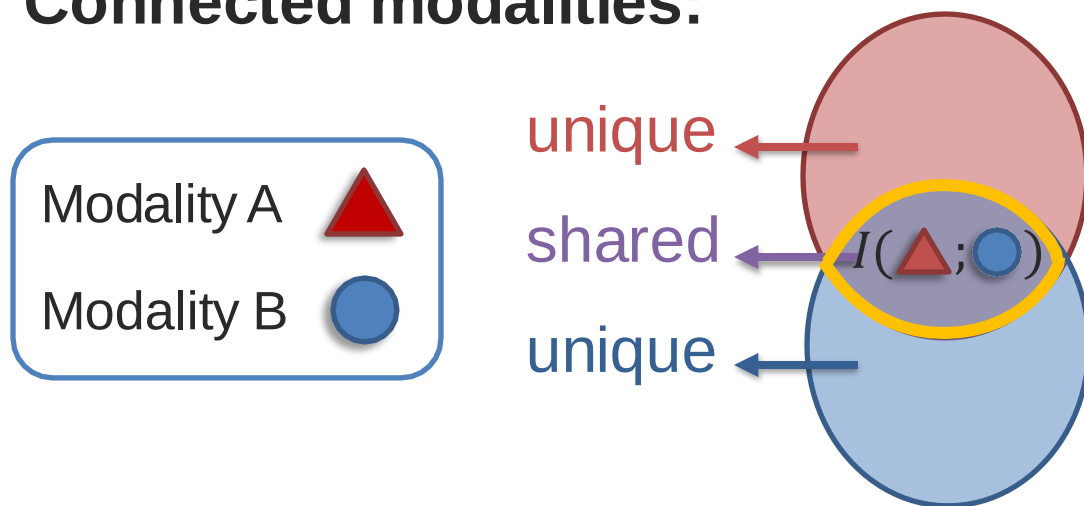


Contrastive Learning and Connected Modalities



$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\text{sim}(\mathbf{z}_A^i, \mathbf{z}_B^i)}{\sum_{j=1}^N \text{sim}(\mathbf{z}_A^i, \mathbf{z}_B^j)}$$

Connected modalities:



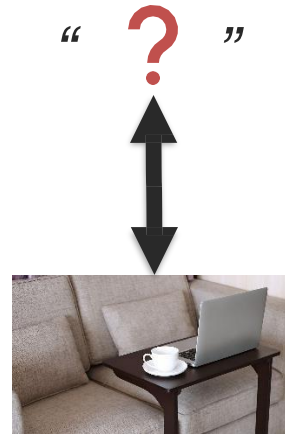
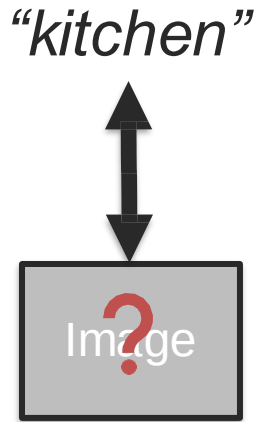
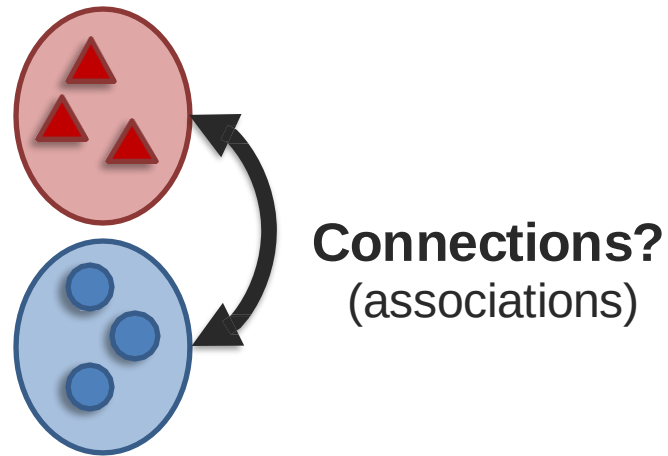
Mutual information $I(X; Y)$

$$\mathbb{E} \left[\log \left(\frac{p(x, y)}{p(x)p(y)} \right) \right]$$



CLIP focuses on shared connections

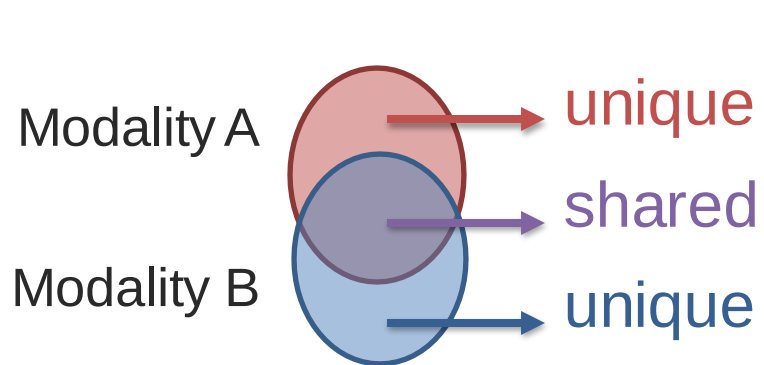
Connections between Modalities



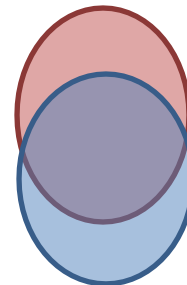
Connection types:

- Co-occurrence
- Correlation
- Causality
- Relationship

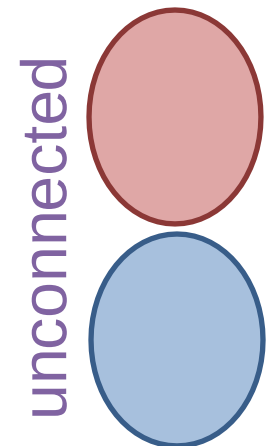
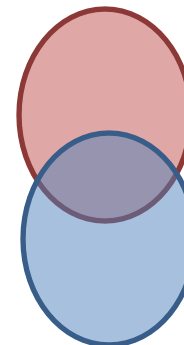
➡ knowledge of one modality provides **information** about the other modality



Strongly
Connected

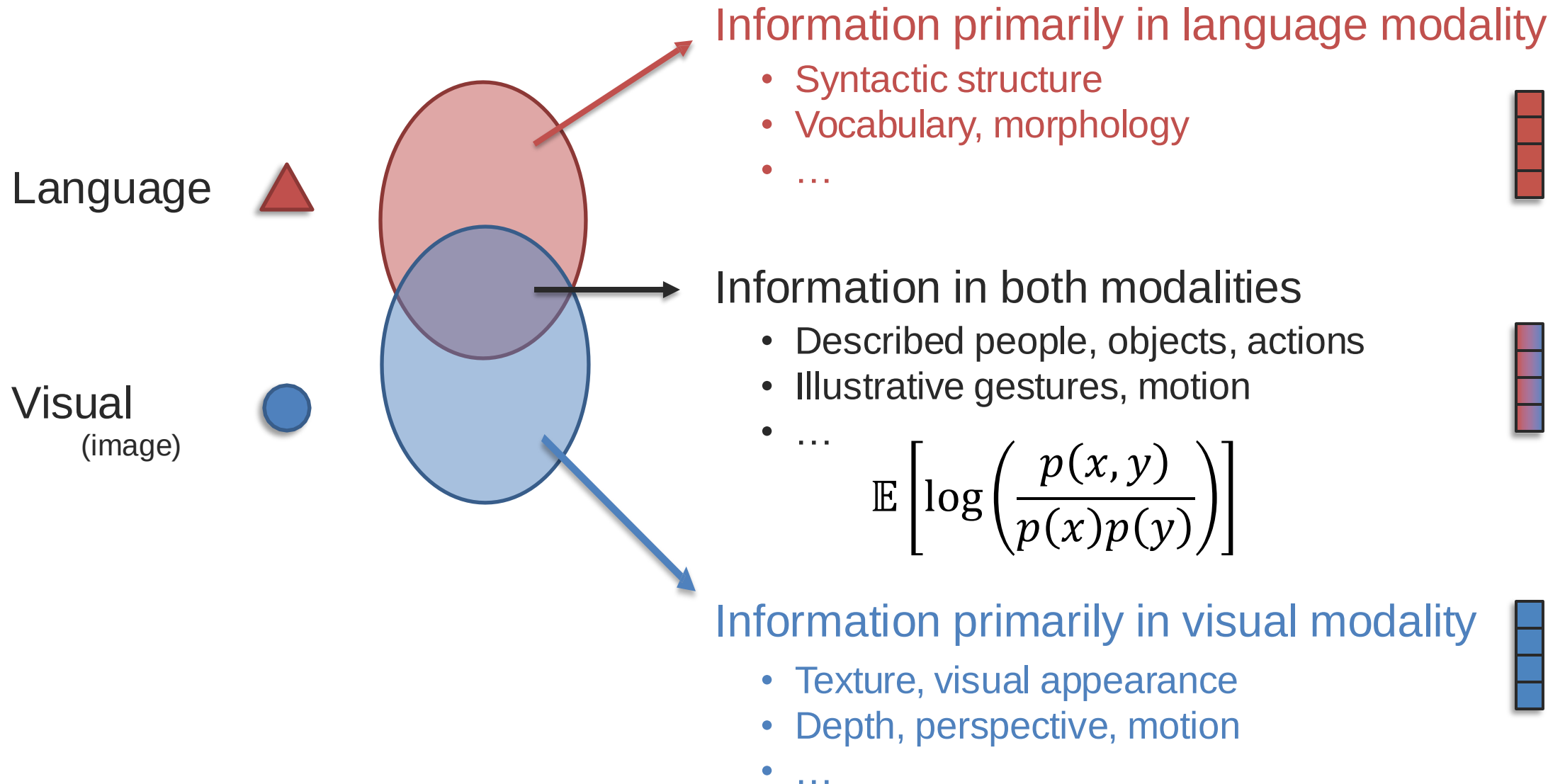


Weakly
Connected



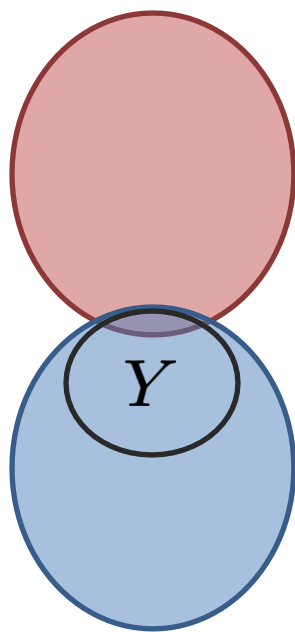
Connections are part of the data... and models will try to learn them.

Modality-Level Factorization

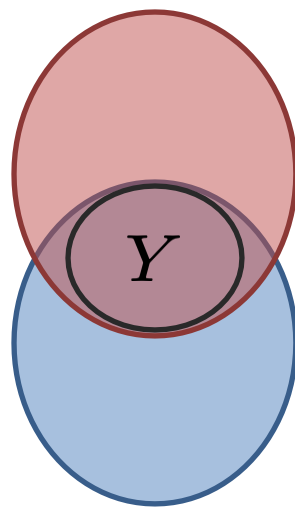


How much information should be shared?

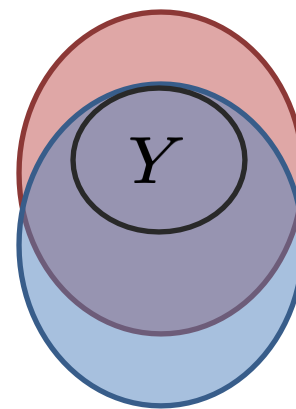
Not enough signal



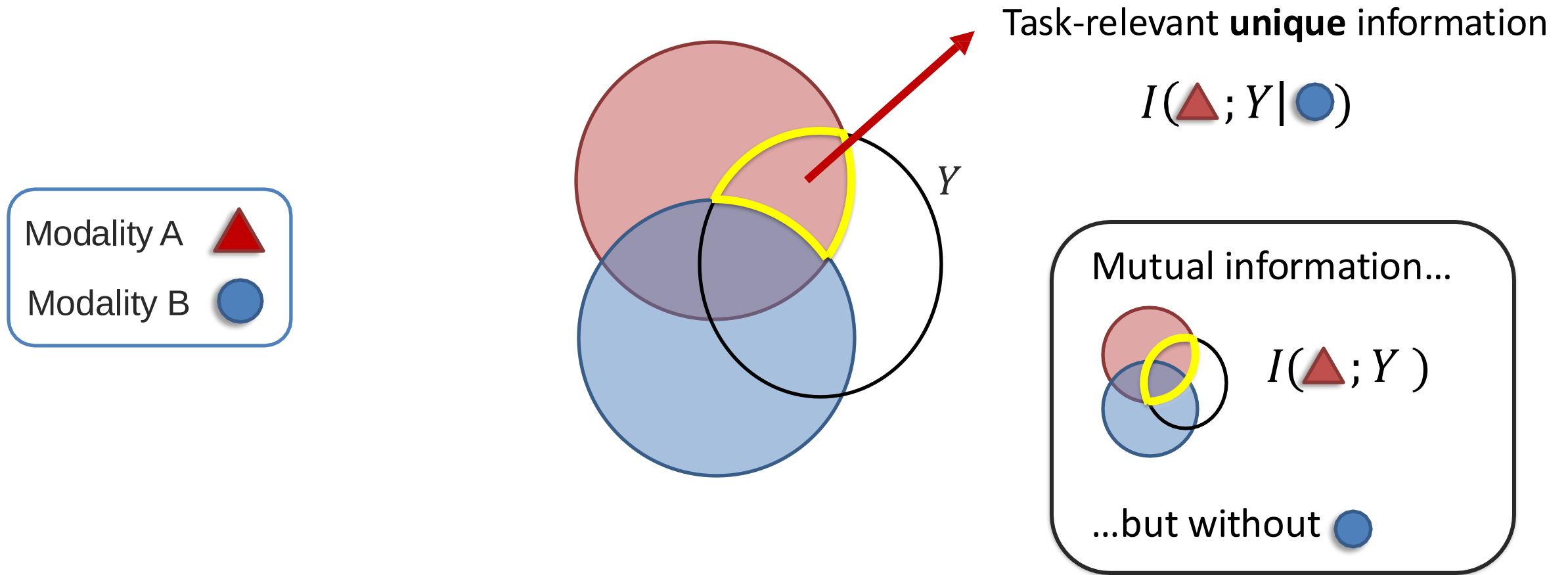
Just right



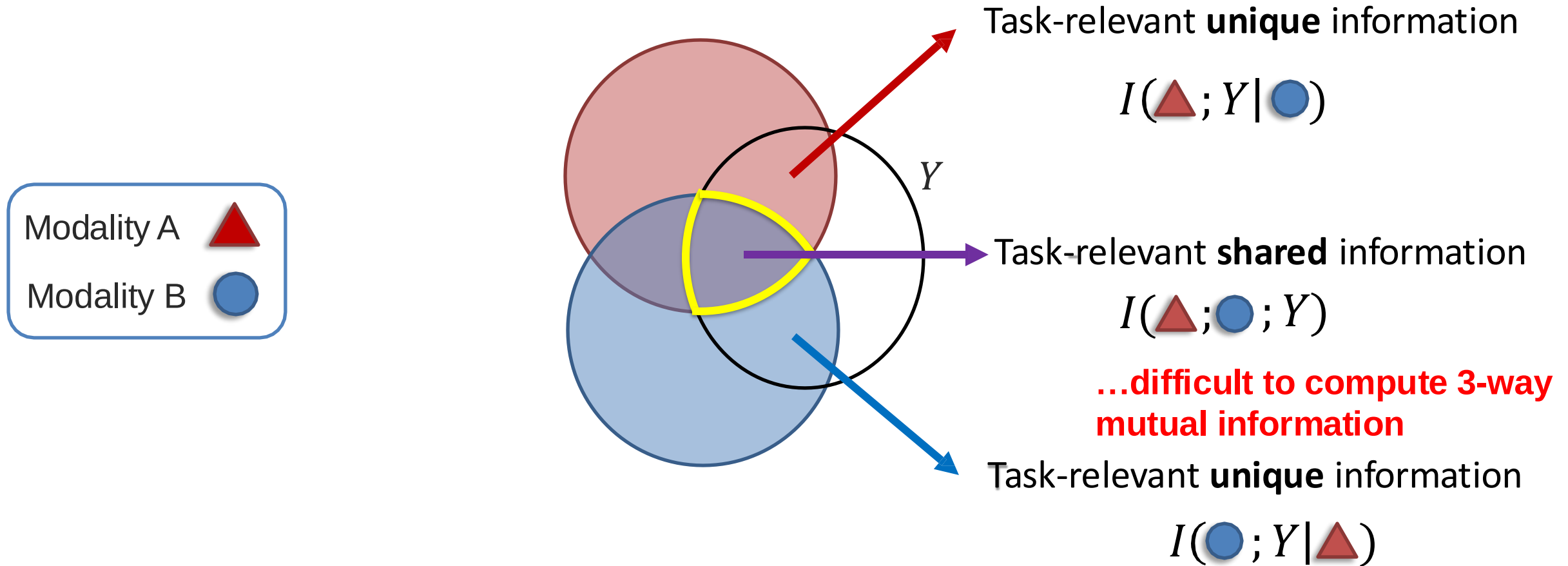
Too much noise



Response and Input Modalities



Response and Input Modalities

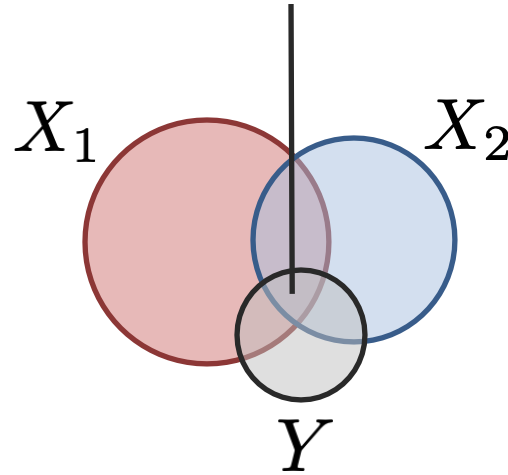


Partial Information Decomposition

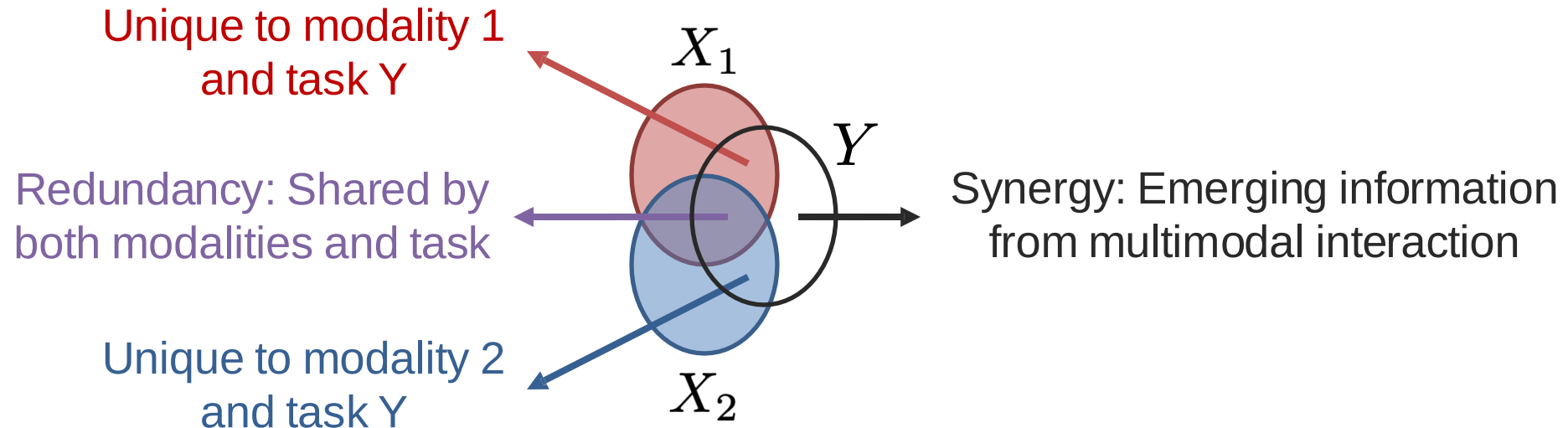
Classical Information Theory

$$I(X_1; X_2; Y) = I(X_1; X_2) - I(X_1; X_2|Y)$$

Can be negative!



Partial Information Decomposition



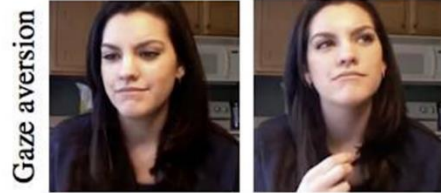
Quantifying Interactions

These interactions can be efficiently estimated – gives a path towards understanding interactions

Language: *And he I don't think he got mad when hah*

I don't know maybe.

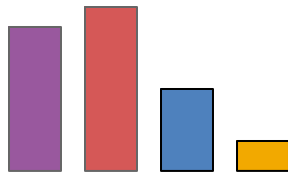
Vision:



Acoustic:

(frustrated voice)

Sentiment



$R \ U_\ell \ U_{av} \ S$

Language/Agreement

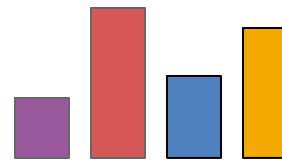
Sheldon :

Its just a *privilege* to watch your mind at work.

- **Text** : suggests a compliment.
- **Audio** : neutral tone.
- **Video** : straight face.

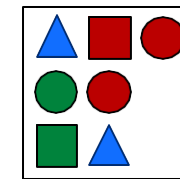


Sarcasm



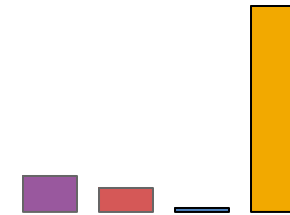
$R \ U_\ell \ U_{av} \ S$

Multimodal Transformer



Is there a red shape above a circle?

VQA



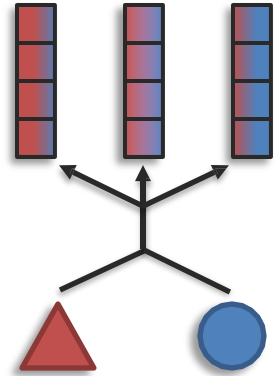
$R \ U_\ell \ U_i \ S$

Multiplicative/Transformer

Also matches human judgment of interactions, and other sanity checks on synthetic datasets

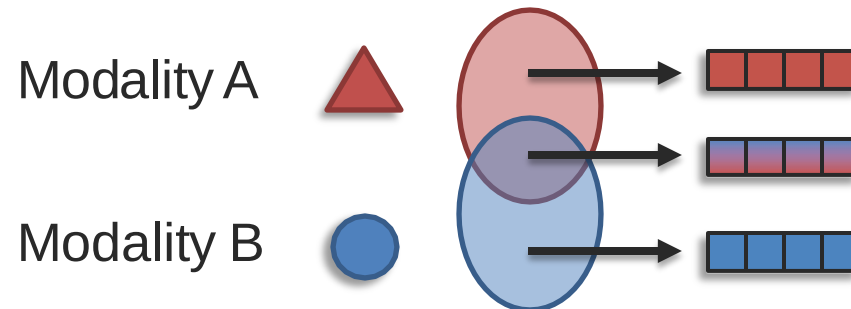
Can also be used to choose most appropriate models – can they be used to better train/design new models?

Representation Factorization

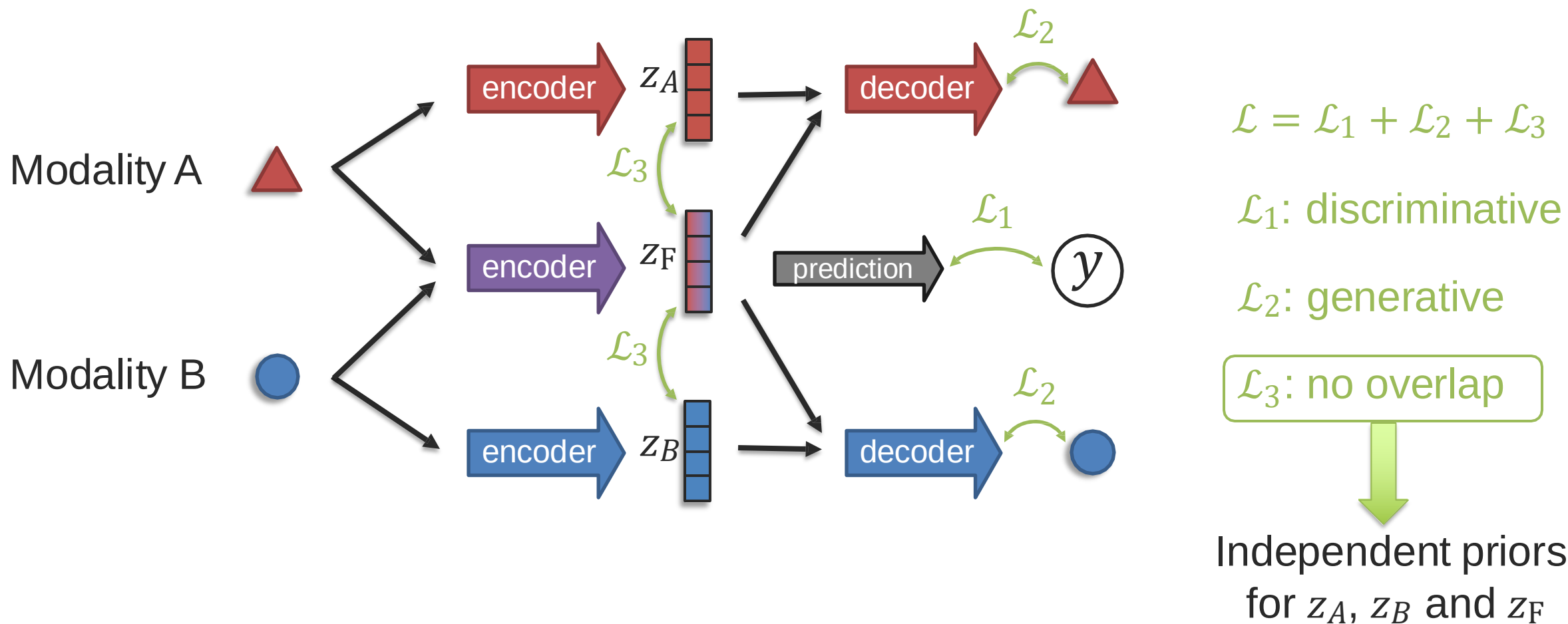


Definition: Learning a new set of representations that reflects multimodal internal structure such as data factorization or clustering

Modality-level Factorization:

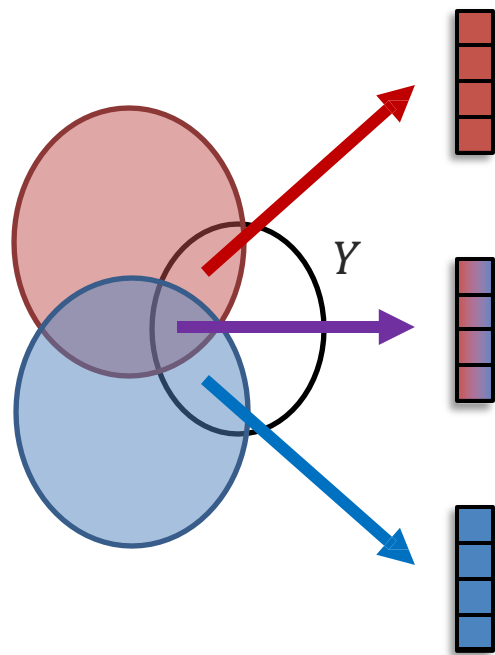


A Generative-Discriminative Approach



Learning Task-relevant Unique Information

Modeling task-relevant unique information



- 2 Maximize task-relevant **unique** information

$$I(\mathbf{Z}; Y | \bullet)$$

- 1 Maximize task-relevant **shared** information

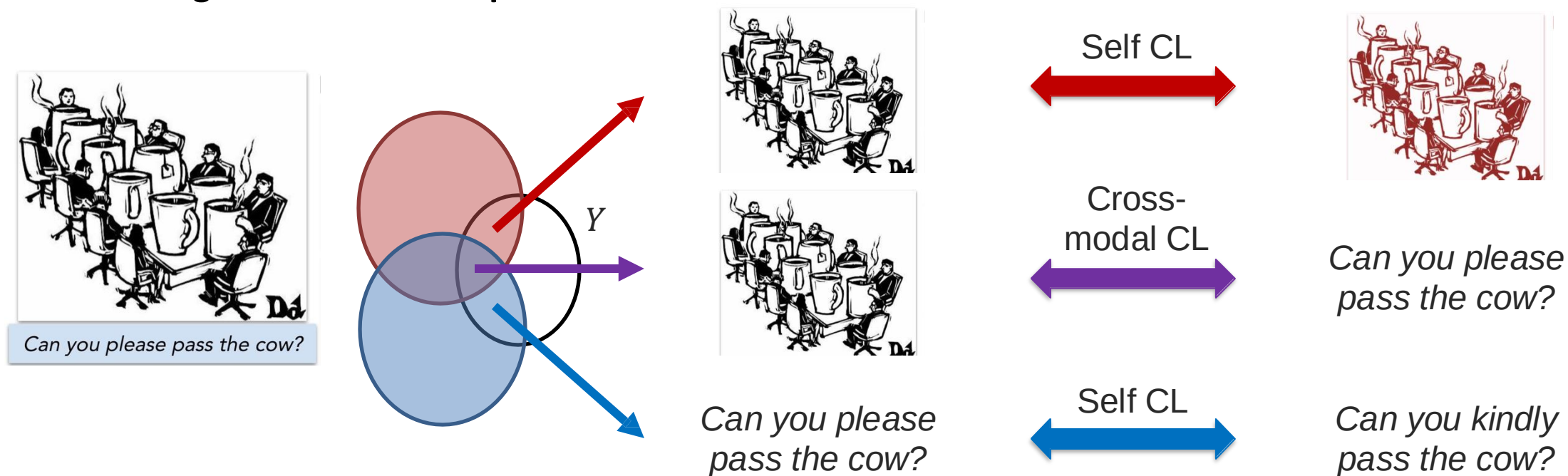
$$I(\mathbf{Z}; \bullet; Y) \quad \text{and} \quad I(\mathbf{Z}; \blacktriangle; Y)$$

- 3 Maximize task-relevant **unique** information

$$I(\mathbf{Z}; Y | \blacktriangle)$$

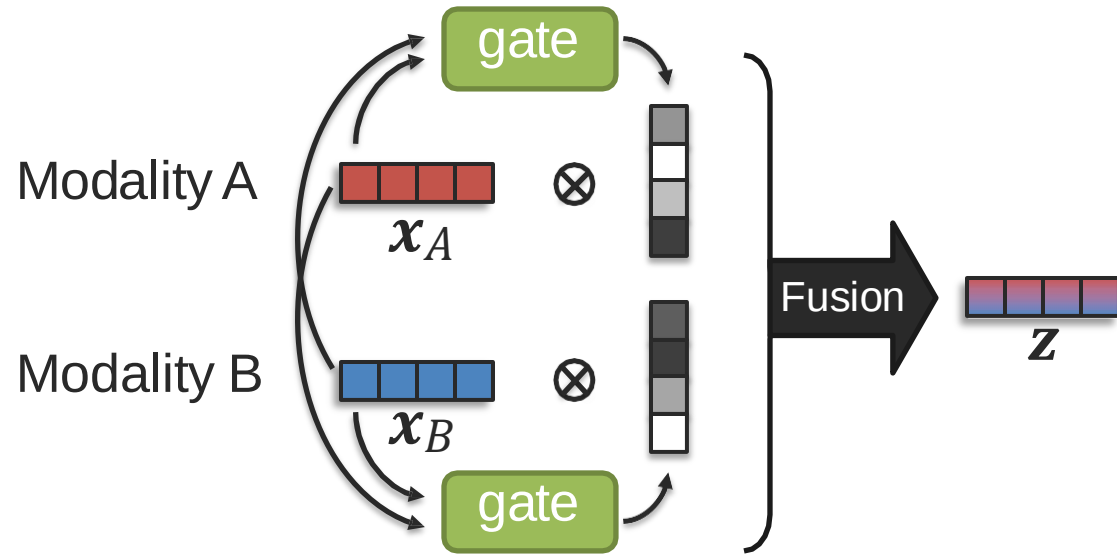
Learning Task-relevant Unique Information

Modeling task-relevant unique information



Approximate task-relevance Y using multi-view data augmentations
New scalable lower and upper bounds on mutual information

Gate Module (Attentive Pooling)



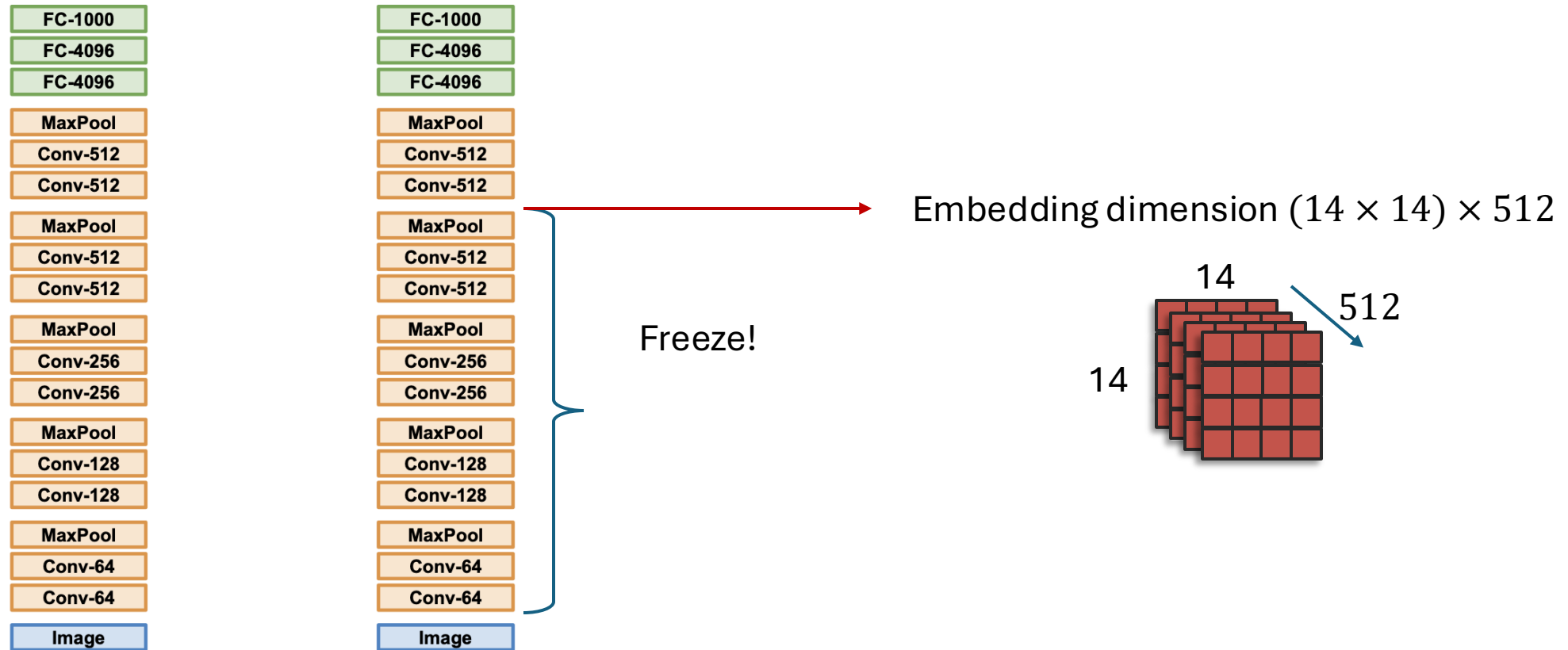
Example with additive fusion:

$$z = g_a(x_A, x_B)x_a + g_b(x_A, x_B)x_b$$

g_A and g_B can be seen as attention functions

Gating is done on the spatial dimension

1. Train on ImageNet 2. Extract features



Attention equation

- Region representation

$$a_I = \sum_{j=1..49} g_I(j) \cdot I_j$$

- Note, the image representation should have spatial dimension

3.2	5.1	9	6	...	0.1	0.4	7.9
-----	-----	---	---	-----	-----	-----	-----

I_j

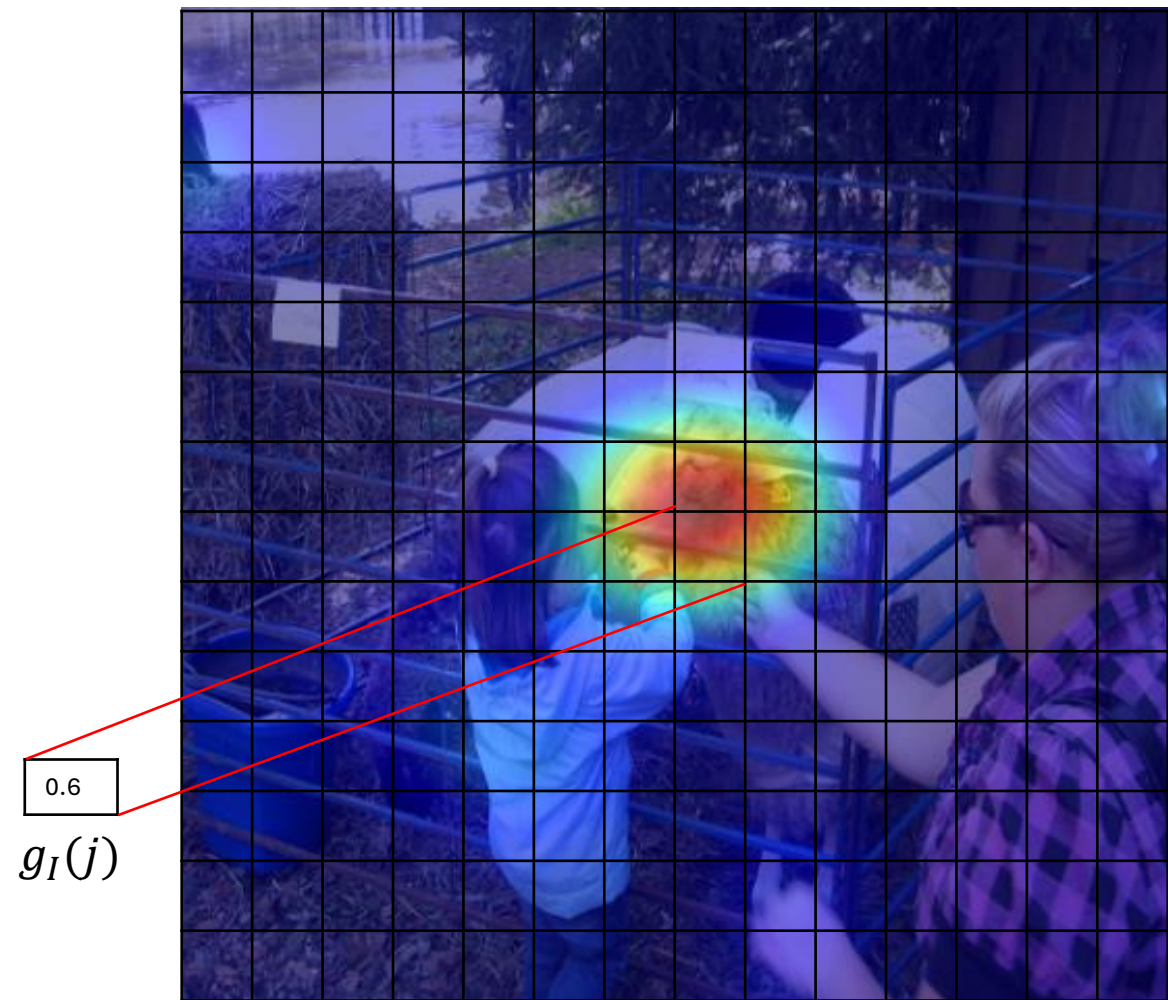


Attention gate equation

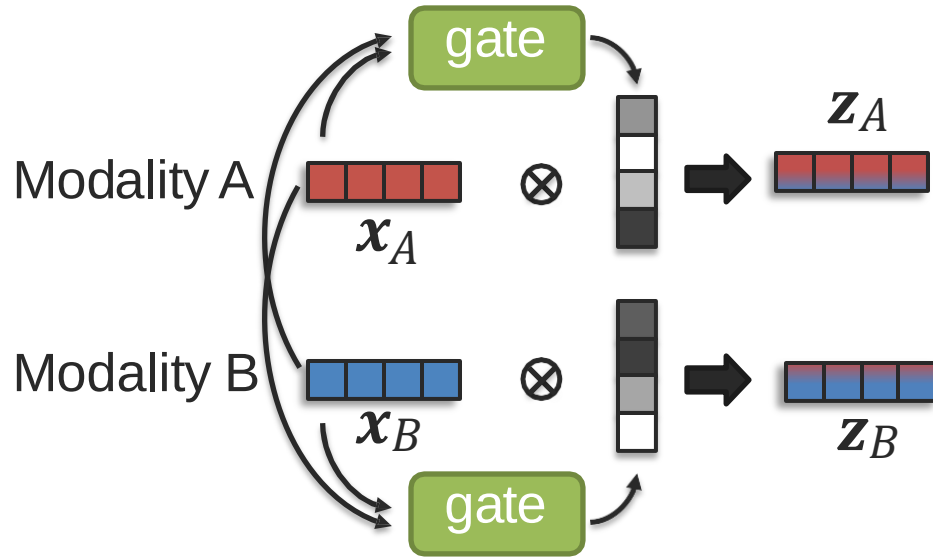
- Image attention:

$$a_I = \sum_{j=1..49} g_I(j) \cdot I_j$$

- How to compute g_I ?
 - E.g., word-guided attention



Gated Coordination



Gated coordination:

$$z_A = g_A(x_A, x_B)x_a$$

$$z_B = g_B(x_A, x_B)x_b$$

Also known as Co-Attention*

*-Hierarchical Question-Image Co-Attention for Visual Question Answering; Lu et al; NIPS'16

Visual question answering



What animal is in the cage?

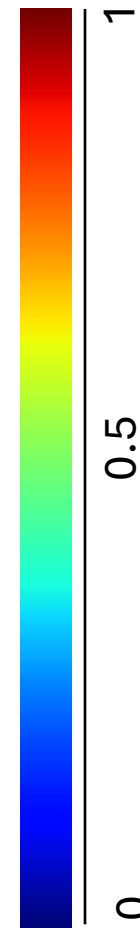
VQA Co-Attention



What **color** is the plastic **container** **inside** the cage?

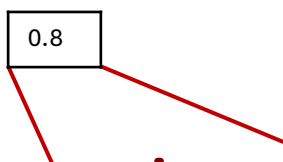


What **animal** is **in the** cage?



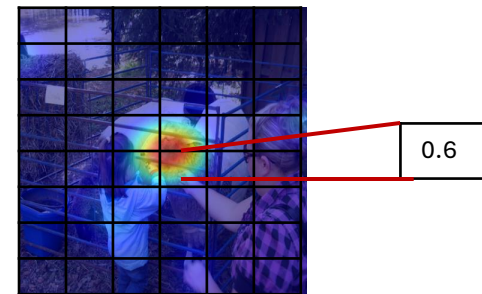
VQA elements

What animal is in the cage?



Elements : words

$$(Q_1, \dots, Q_{15}) \in \mathbb{R}^{15 \times d_Q}$$



Elements: image regions

$$(I_1, \dots, I_{49}) \in \mathbb{R}^{49 \times d_I}$$

VQA Co-Attention

- Given an image I and a question Q
- We infer the amount of attention that should be given to each image-region and a word by computing attention belief b :

$$g_I(i) \propto \exp(b(I, Q, i)), \quad g_Q(i) \propto \exp(b(I, Q, i))$$

- We calculate the attend representation:

$$a_Q = \sum_{i=1..15} g_Q(i) \cdot Q_j \quad ; \quad a_I = \sum_{j=1..49} g_I(i) \cdot I_i$$

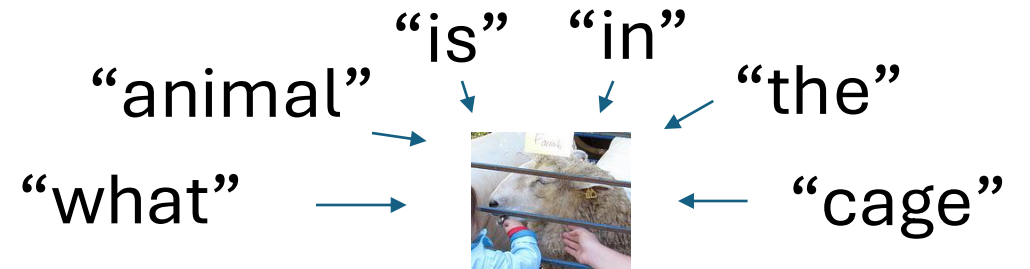
Factorize the attention belief

$$b_I(I, Q, i) \propto \exp(w_I \psi_I(i) + w_{I,Q} \mu_{Q \rightarrow I}(i))$$

Unary ($\psi_I(i)$): region information

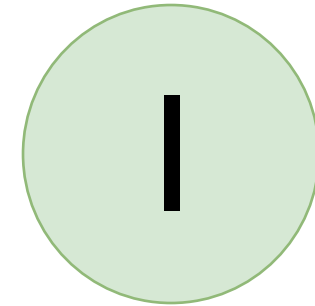
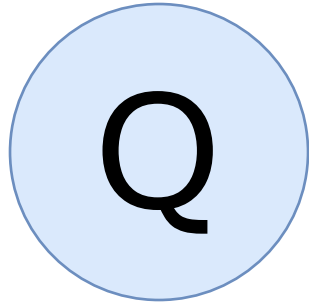


question-to-Image ($\mu_{Q \rightarrow I}(i)$):
interactions of regions with words



Factor Graph Attention

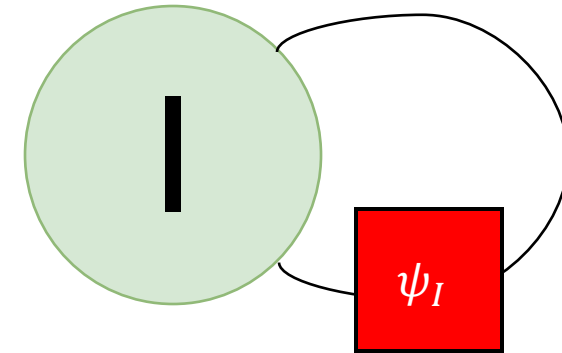
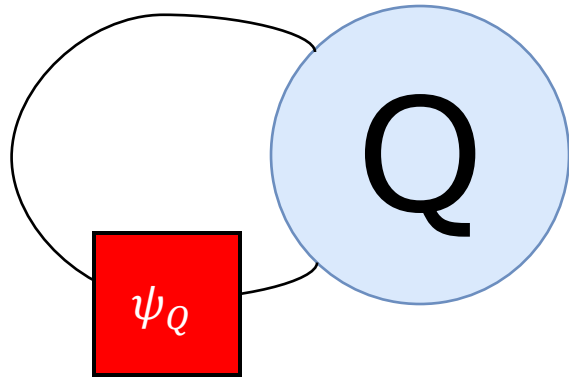
- Each modality is treated as a node



What animal is in the cage?



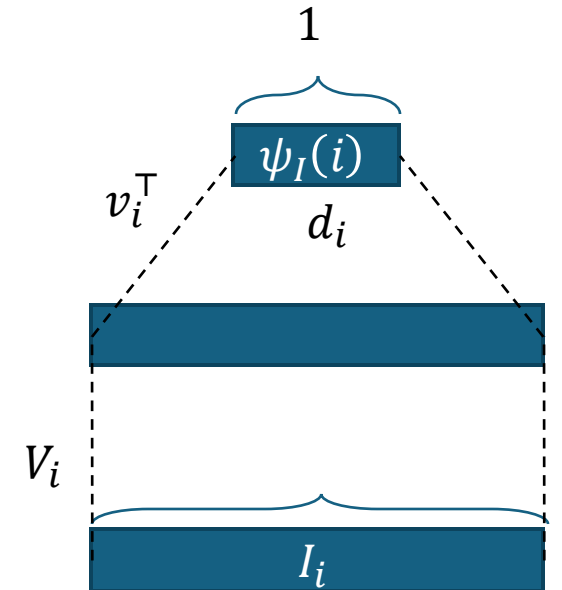
Unary factor



For each image region:

$$\psi_I(i) = v_i^\top \text{relu}(V_i I_i)$$

Where weights $v_i^\top \in \mathbb{R}^{d_I \times 1}$ and $V_i \in \mathbb{R}^{d_I \times d_I}$



$$b_I(I, Q, i) \propto \exp(w_I \psi_I(i) + w_{I,Q} \mu_{Q \rightarrow I}(i))$$

Unary attention visualize

- Entity information reacts to strong features across the dataset
- Can reveal dataset **bias**



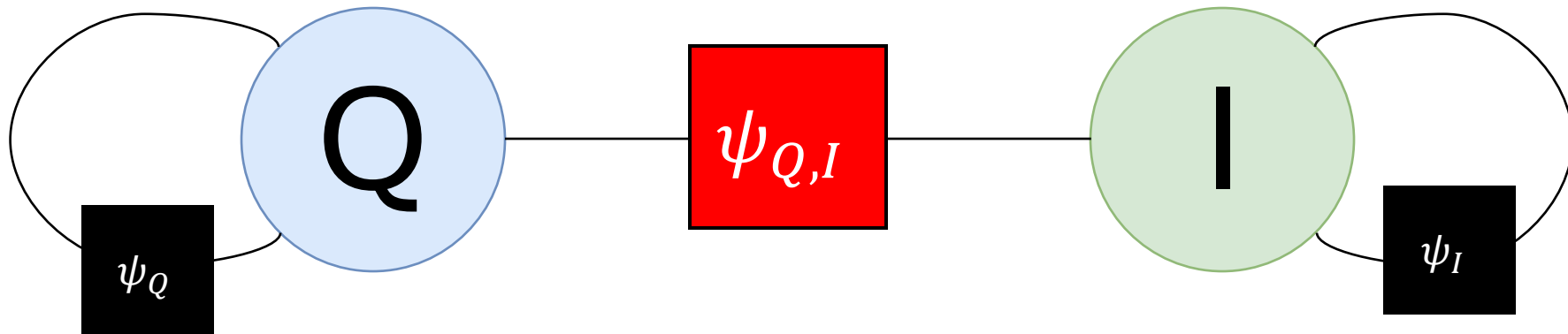
What animal is in the cage?



What color is the plastic container inside the cage?

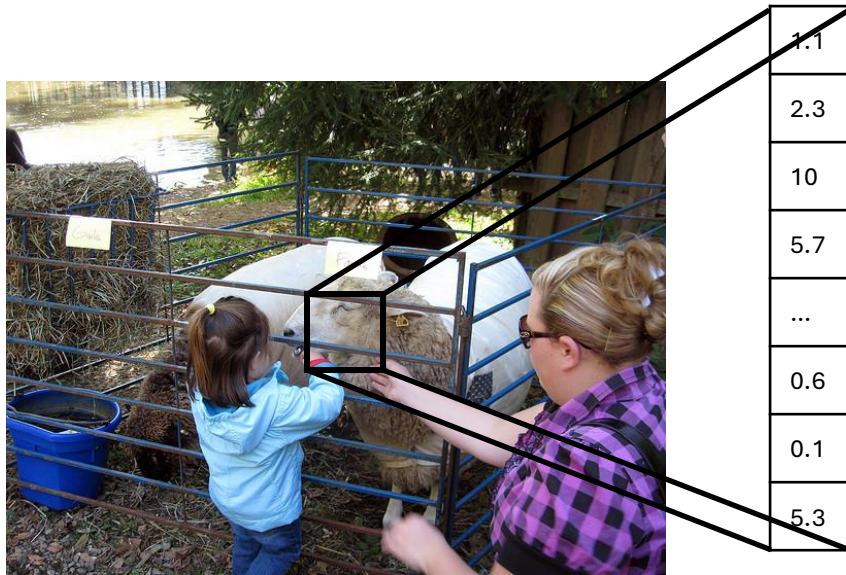
$$b_I(I, Q, i) \propto \exp(w_I \psi_I(i) + w_{I,Q} \mu_{Q \rightarrow I}(i))$$

Pairwise Interaction factor



$$b_I(I, Q, i) \propto \exp(w_I \psi_I(i) + w_{I,Q} \mu_{Q \rightarrow I}(i))$$

Interactions via dot-product



•

3.2	5.1	9	6	...	0.1	0.4	7.8
-----	-----	---	---	-----	-----	-----	-----

 = 6.5

What animal is in the cage?

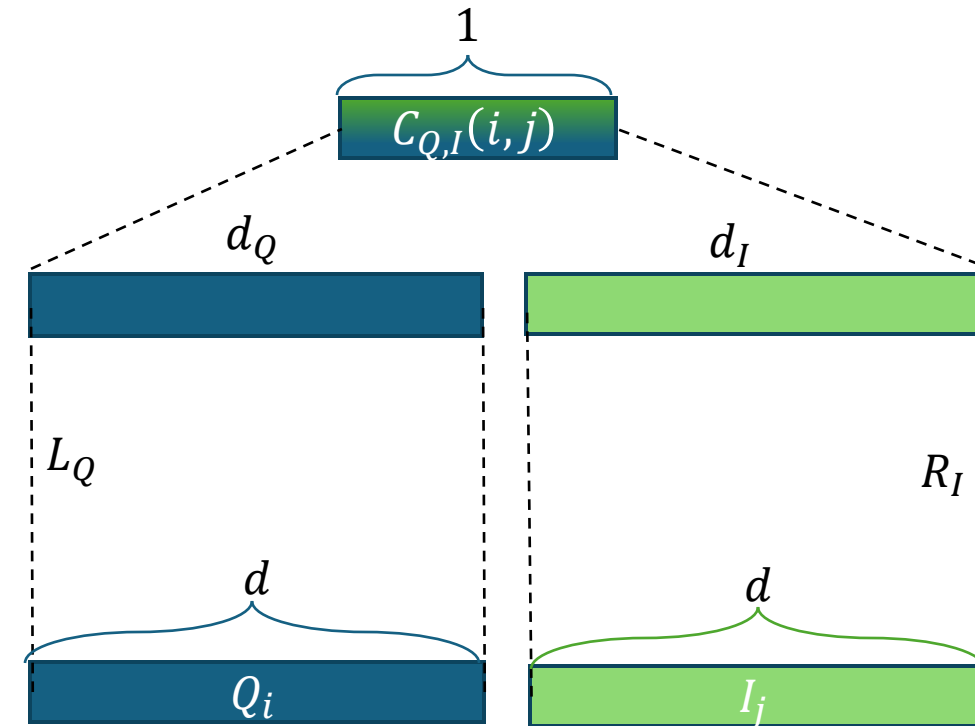
$$b_I(I, Q, i) \propto \exp(w_I \psi_I(i) + w_{I,Q} \mu_{Q \rightarrow I}(i))$$

The interaction is learned

- For each two modalities

$$C_{Q,I}(i,j) = \left(\frac{L_Q Q_i}{\|L_Q Q_i\|} \cdot \frac{R_I I_j}{\|R_I I_j\|} \right)$$

- Where weights $L_q \in \mathbb{R}^{d_Q \times d}$ and $R_i \in \mathbb{R}^{d_I \times d}$



$$b_I(I, Q, i) \propto \exp(w_I \psi_I(i) + w_{I,Q} \mu_{Q \rightarrow I}(i))$$

Pairwise interaction matrix



What animal
is in the cage?

$$C_{Q,I}(i,j) = \left(\frac{L_Q Q_i}{\|L_Q Q_i\|} \cdot \frac{R_I I_j}{\|R_I I_j\|} \right)$$

			
What	1.2	2.3	1.7
animal	-0.4	6.5	4.2
is	1.3	1.2	0.6

$$b_I(I, Q, i) \propto \exp(w_I \psi_I(i) + w_{I,Q} \mu_{Q \rightarrow I}(i))$$

Message from question to image

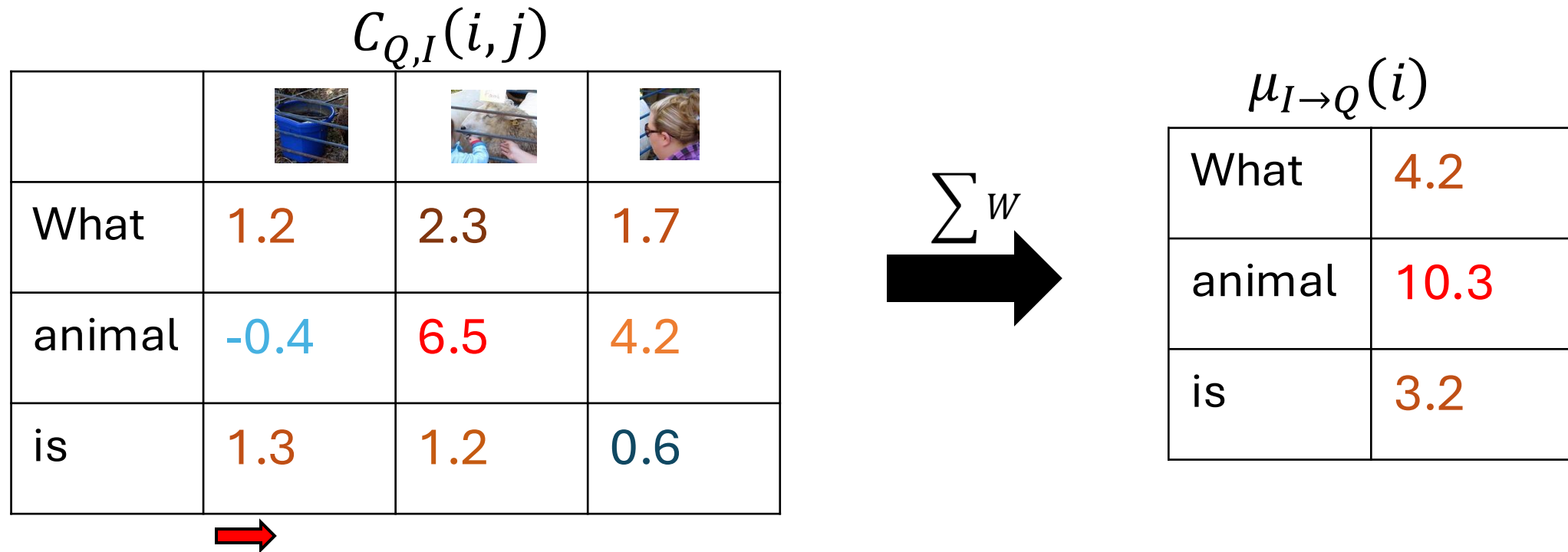
- Messages via weighted marginalization: $\mu_{Q \rightarrow I}(i) = \sum_{Q_j} W_{I,i,j} C_{Q,I}(i,j)$

	$C_{Q,I}(i,j)$				$\mu_{Q \rightarrow I}(i)$	
						2.1
What	1.2	2.3	1.7	$\sum^w$ 		9
animal	-0.4	6.5	4.2			6.5
is	1.3	1.2	0.6			

$$b_I(I, Q, i) \propto \exp(w_I \psi_I(i) + w_{I,Q} \mu_{Q \rightarrow I}(i))$$

Message from image to question

- Messages via weighted marginalization: $\mu_{Q \rightarrow I}(i) = \sum_{I_j} W_{I,i,j} C_{Q,I}(j, i)$



$$b_I(I, Q, i) \propto \exp(w_I \psi_I(i) + w_{I,Q} \mu_{Q \rightarrow I}(i))$$

Message's visualization

- Messages reacts to highly word-correlated regions



What **animal** is in the cage?

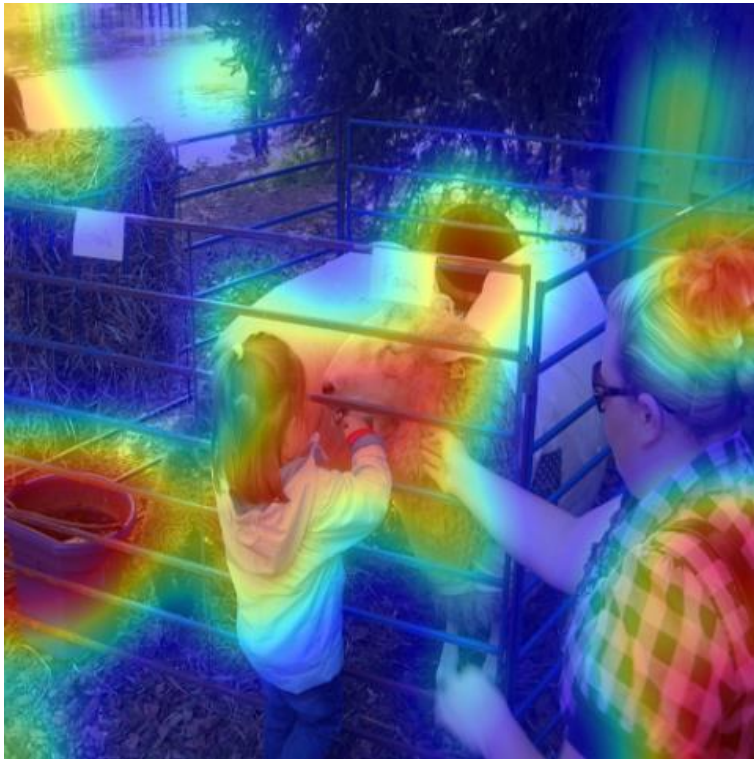


What **color** is the **plastic container** inside the cage?

$$b_I(I, Q, i) \propto \exp(w_I \psi_I(i) + w_{I,Q} \mu_{Q \rightarrow I}(i))$$

Unary potential, pairwise messages and belief

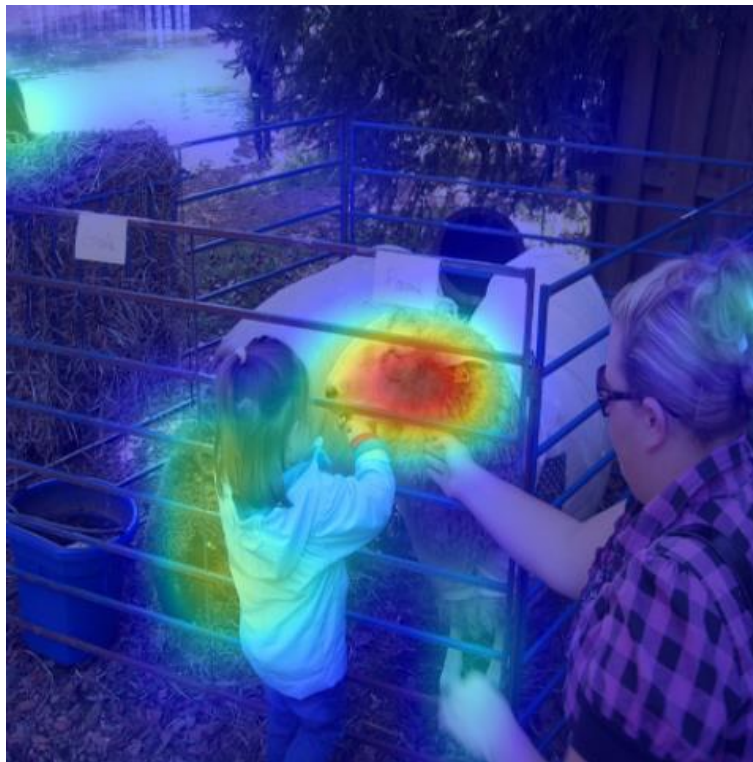
$$\psi_I(i)$$



What **animal** is in the cage?

$$\psi_Q(i)$$

$$\mu_{Q \rightarrow I}(i, j)$$



What **animal** is in the cage?

$$\mu_{I \rightarrow Q}(i, j)$$

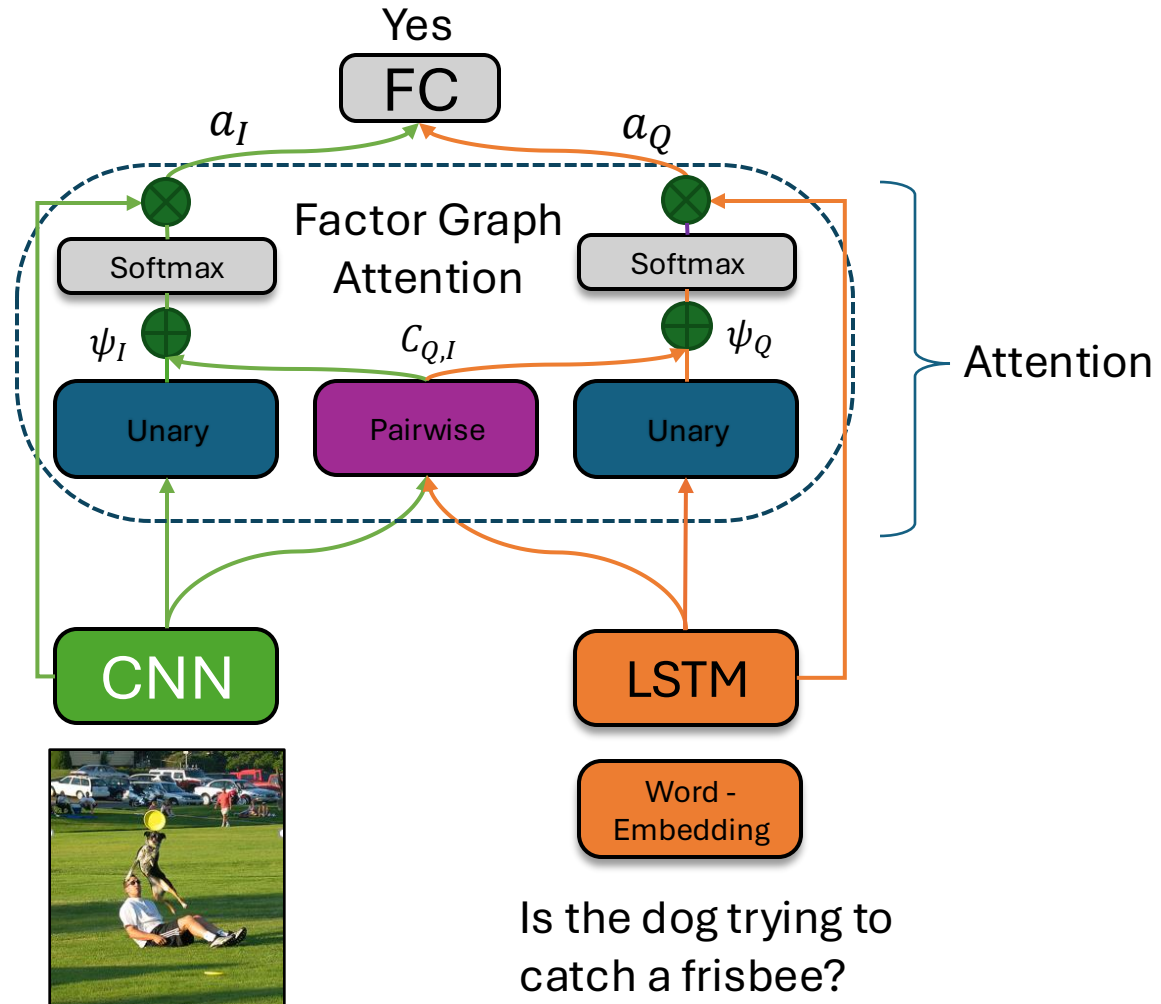
$$b_I(i)$$



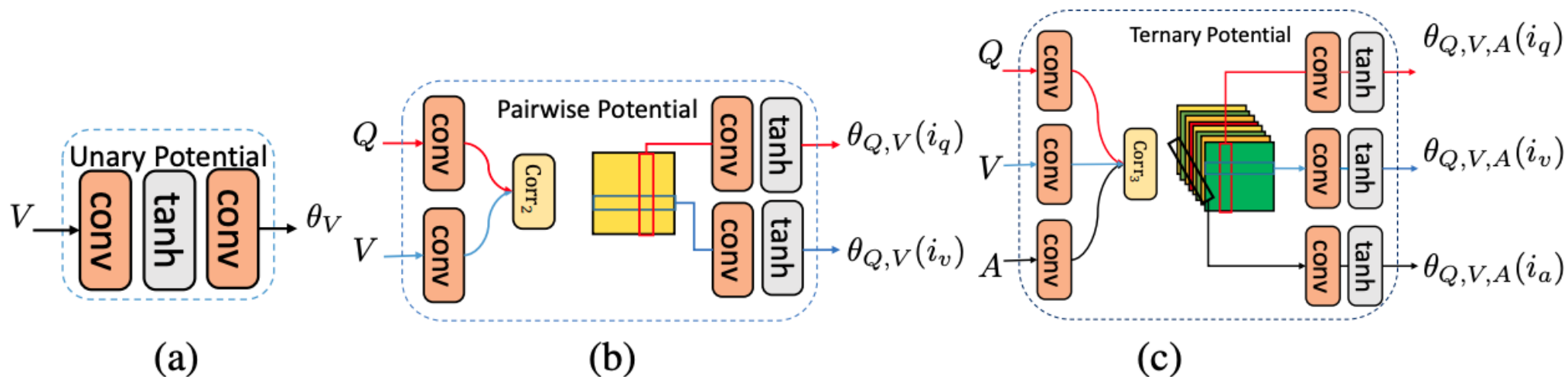
What **animal** is in the cage?

$$b_Q(j)$$

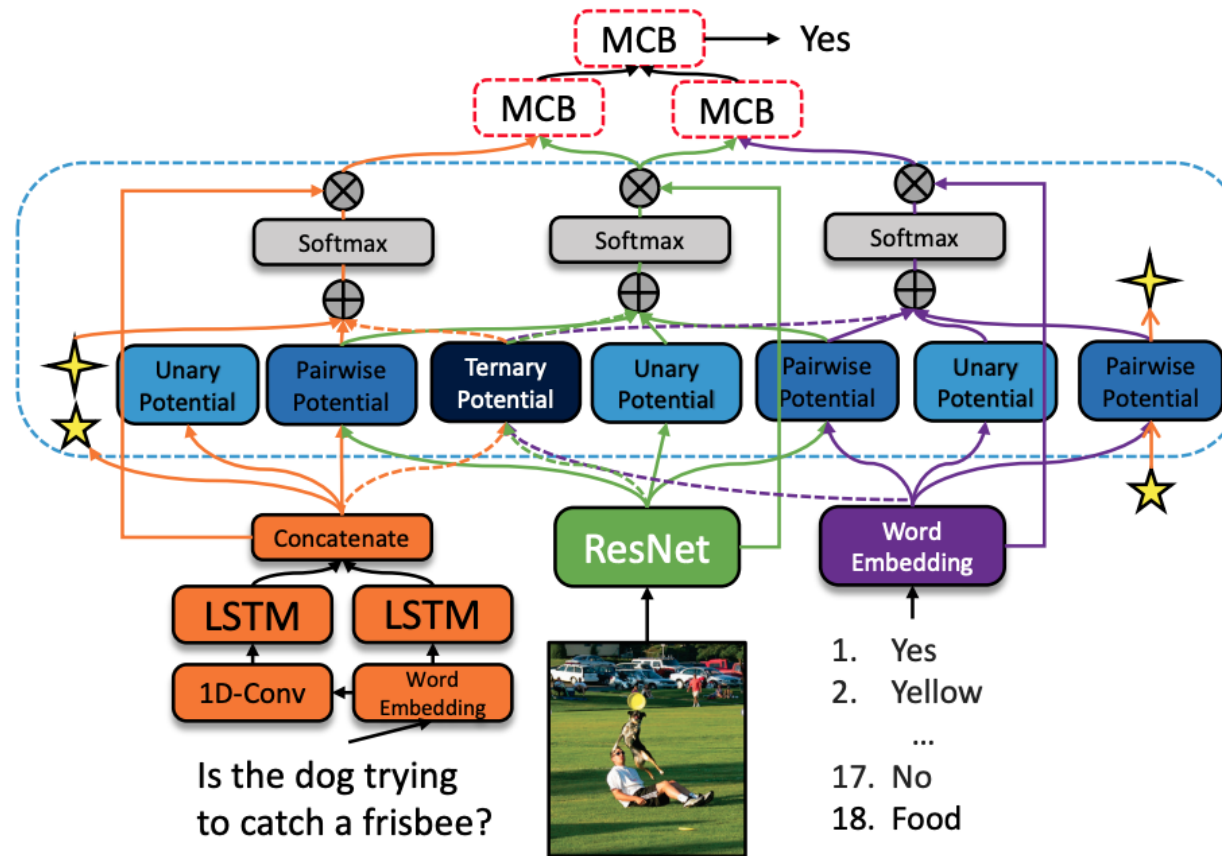
Strong VQA baseline



High-order attention belief



State-of-the-art MC-VQA



Many utilities: Visual Dialog

<i>Image</i>	<i>Caption</i>	<i>Answers</i>
	a group of kids around a table with a cake .	1. Green 2. Yes 3. Not Sure 4. No 5. Yellow 6. 5 Candles ... 99. Five 100. Chocklate
	<i>History</i> Q: How many kids? A: 5 Q: Is this birthday party? A: Yes Q: Any adults? A: lady lighting candle	
<i>Question</i> Is this outdoors?		

Visual Dialog – Many gate attentions

Question

are kids wearing hats ?

History

Q: how many kids ?

A: 5

Q: is this birthday party?

A: yes

Q: any adults ?

A: lady lighting candle

Q: is this outdoors ?

A: not very sure

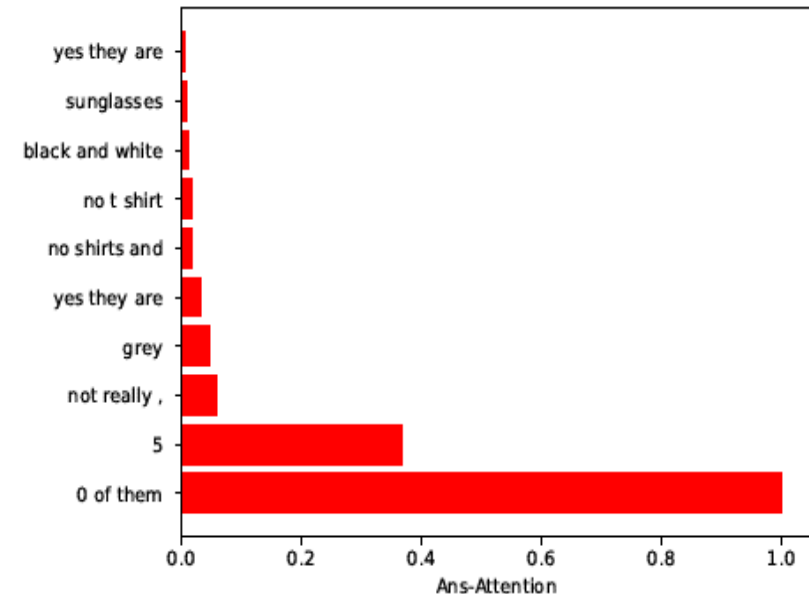
Q: is table round ?

A: no



Caption

a group of kids around a table with a cake .



All gates depends on each other

Question

does cake have candles ?

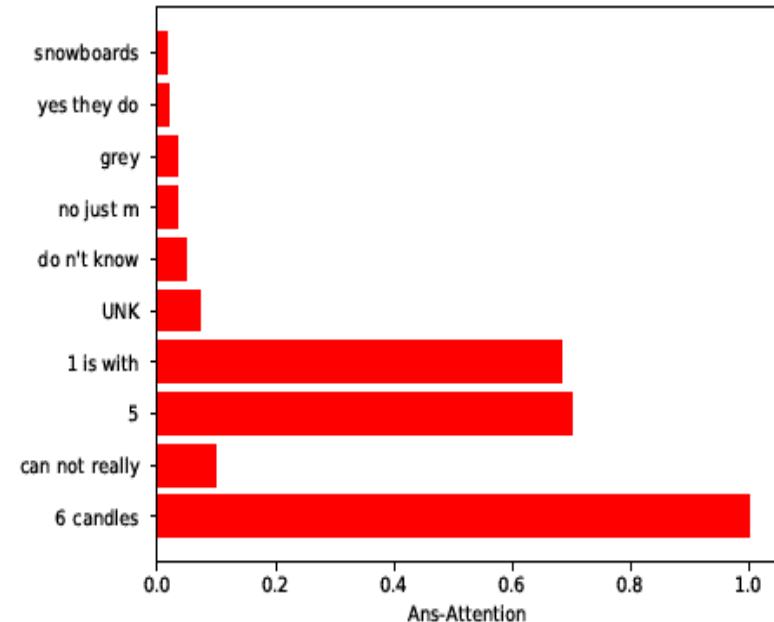
History

Q: how many kids ?
A: 5
Q: is this birthday party ?
A: yes
Q: any adults ?
A: lady lighting candle
Q: is this outdoors ?
A: not very sure
Q: is table round ?
A: no
Q: are kids wearing hats ?
A: 0 of them wearing hats



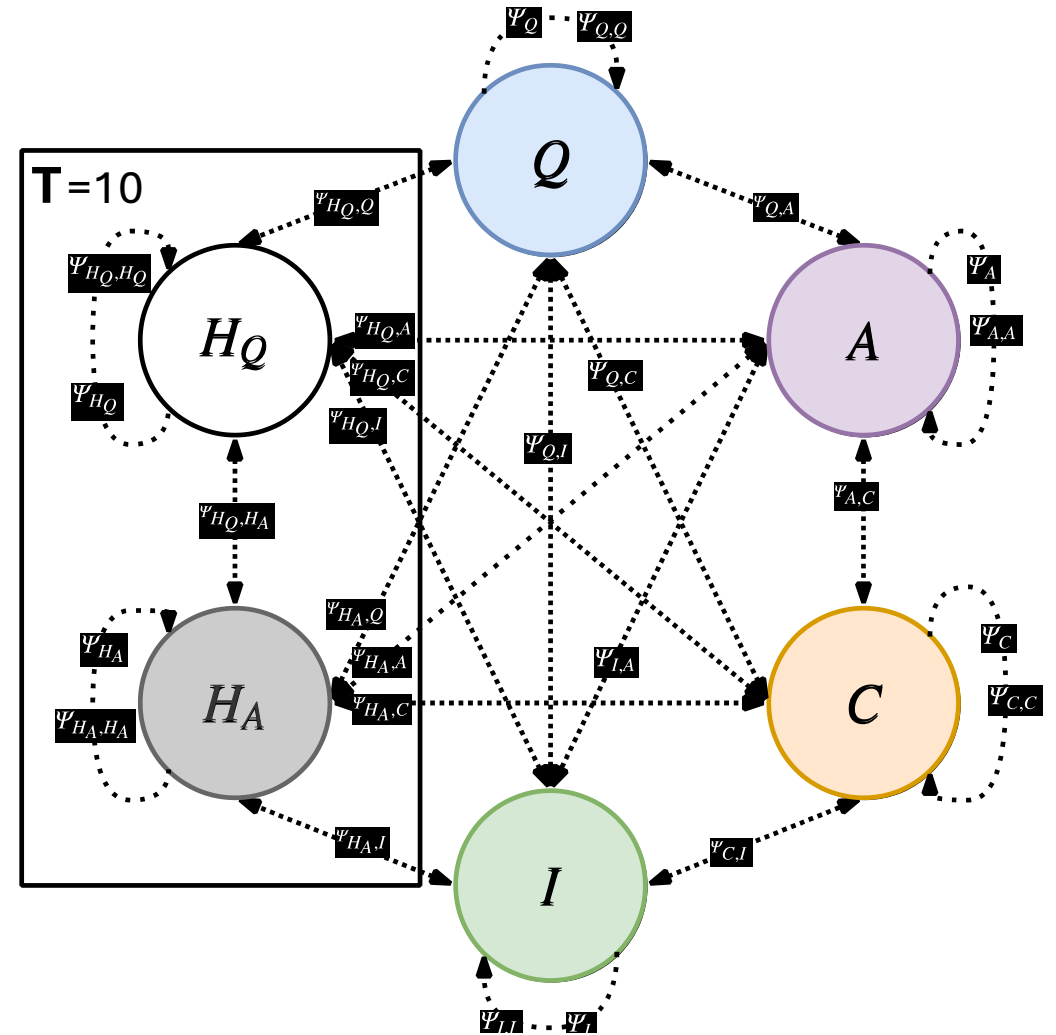
Caption

a group of kids around a table with a cake .



Factor Graph Attention

- **Node** - attention distribution over utility's elements
- **Factors** (i.e., Potentials) – model information and interactions between entities
- Aggregate information from utilities, which are neighbors in the graph



Any number of modalities: Messages and beliefs

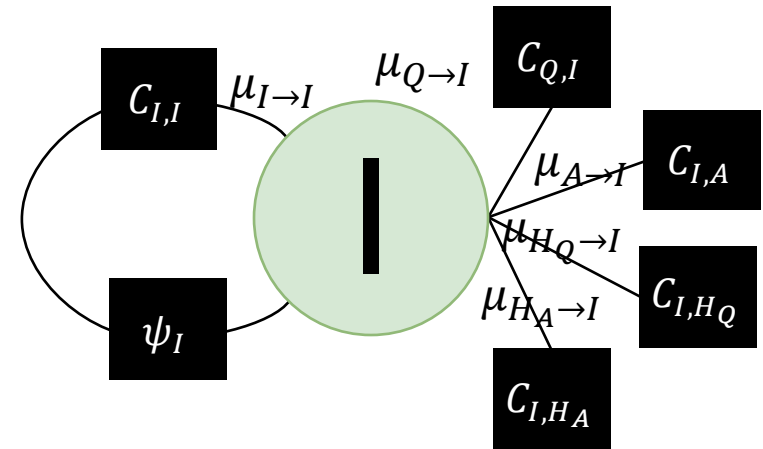
- We collect dependencies via

$$\mu_{Q \rightarrow I}(i) = \sum_{Q_j} W_{i,j} C_{Q,I}(i,j)$$

- We infer the amount of attention that should be given to each entity

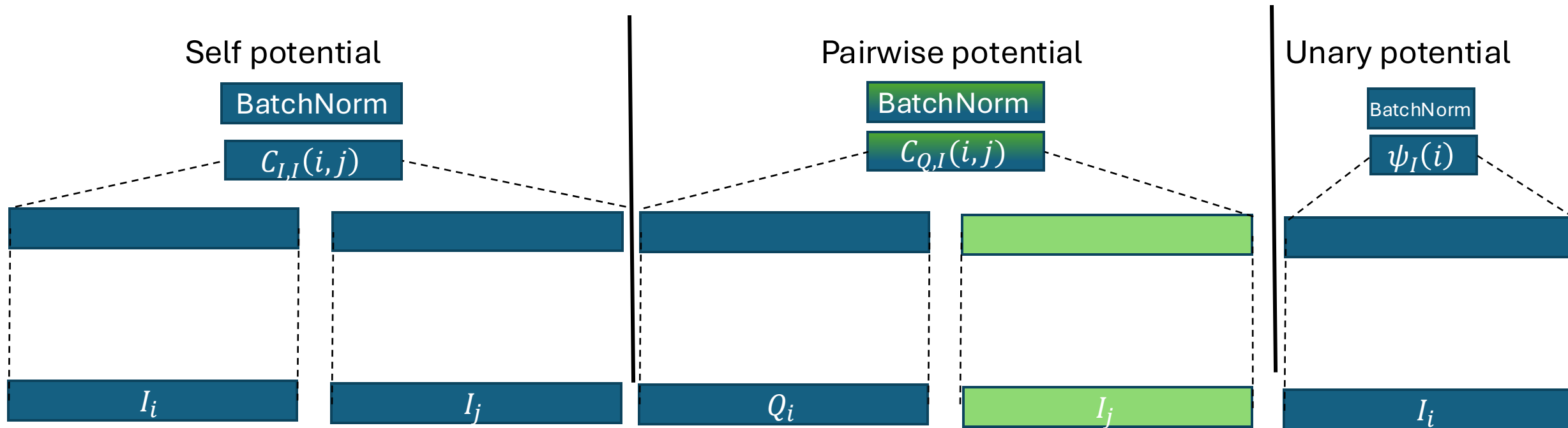
$$b_I(i) \propto \exp \left(w_I \psi_I(i) + \sum_{U \in \mathcal{U}} w_{I,U} \mu_{U \rightarrow I}(i) \right)$$

↑
set of modalities



Factors normalization

- We want to avoid situations where pairwise scores (e.g., image and question) bias another one (e.g., image and caption)
- we use a **batch normalization** for each factor output

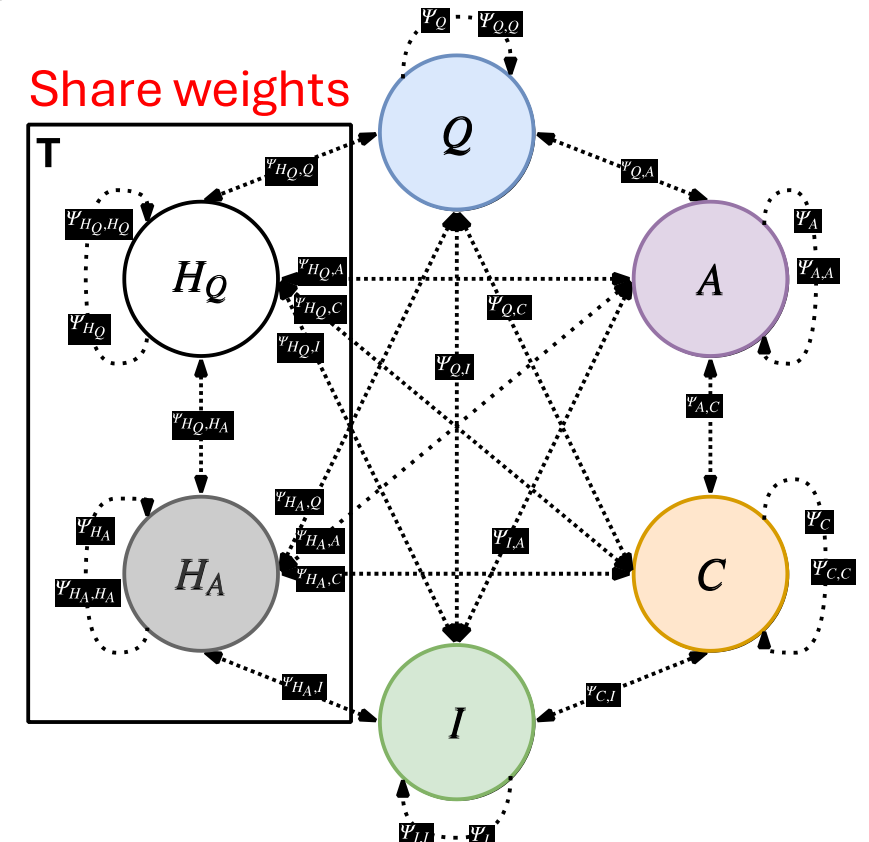


Weight sharing for similar interactions

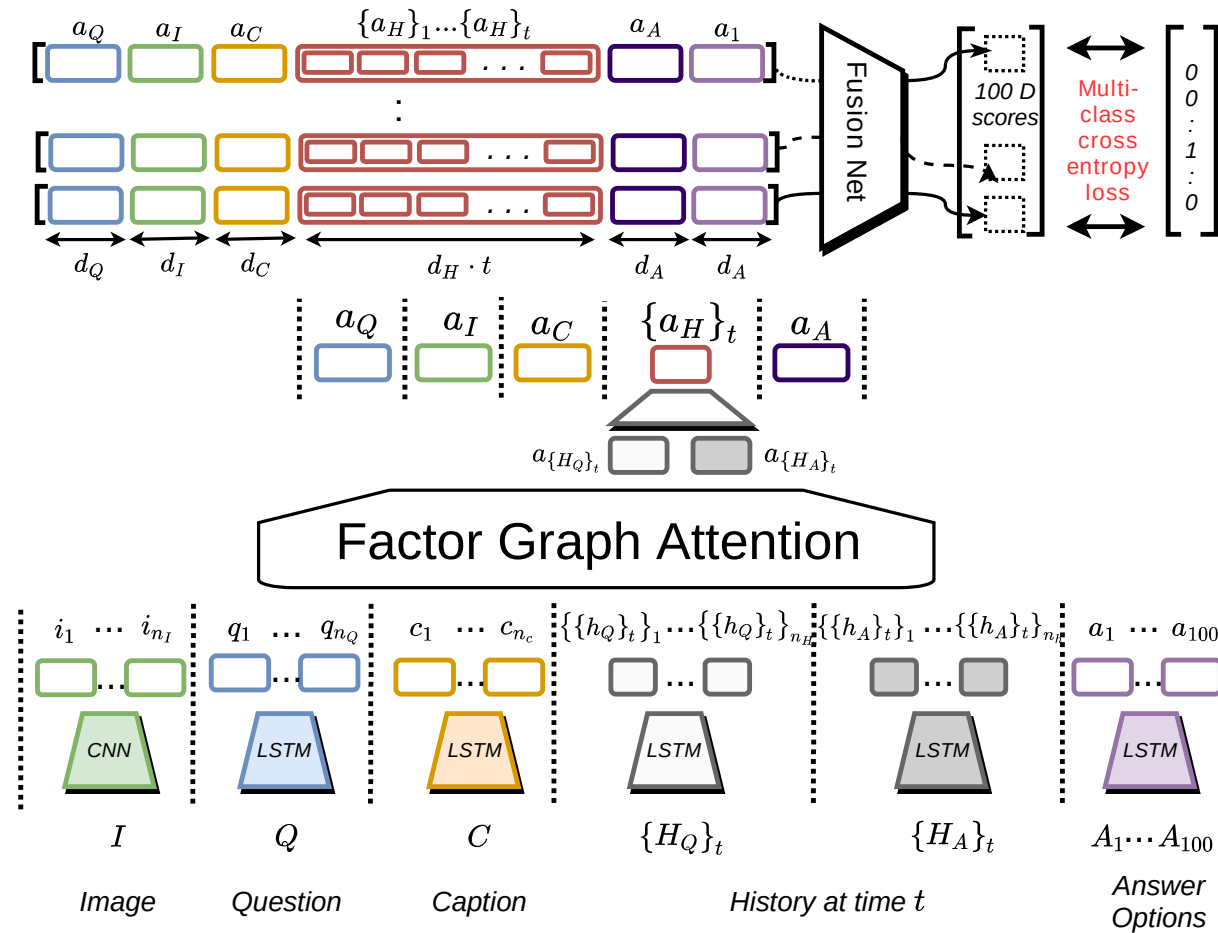
- In case of similar utilities, we can use the same interaction factor
 - e.g., History-Questions, and History-Answers

Reminder, pairwise interaction:

$$C_{A,B}(i,j) = \left(\frac{L_A A_i}{\|L_A A_i\|} \right)^\top \cdot \frac{R_B B_j}{\|R_B B_j\|}$$



State-of-the-art Visual Dialog architecture



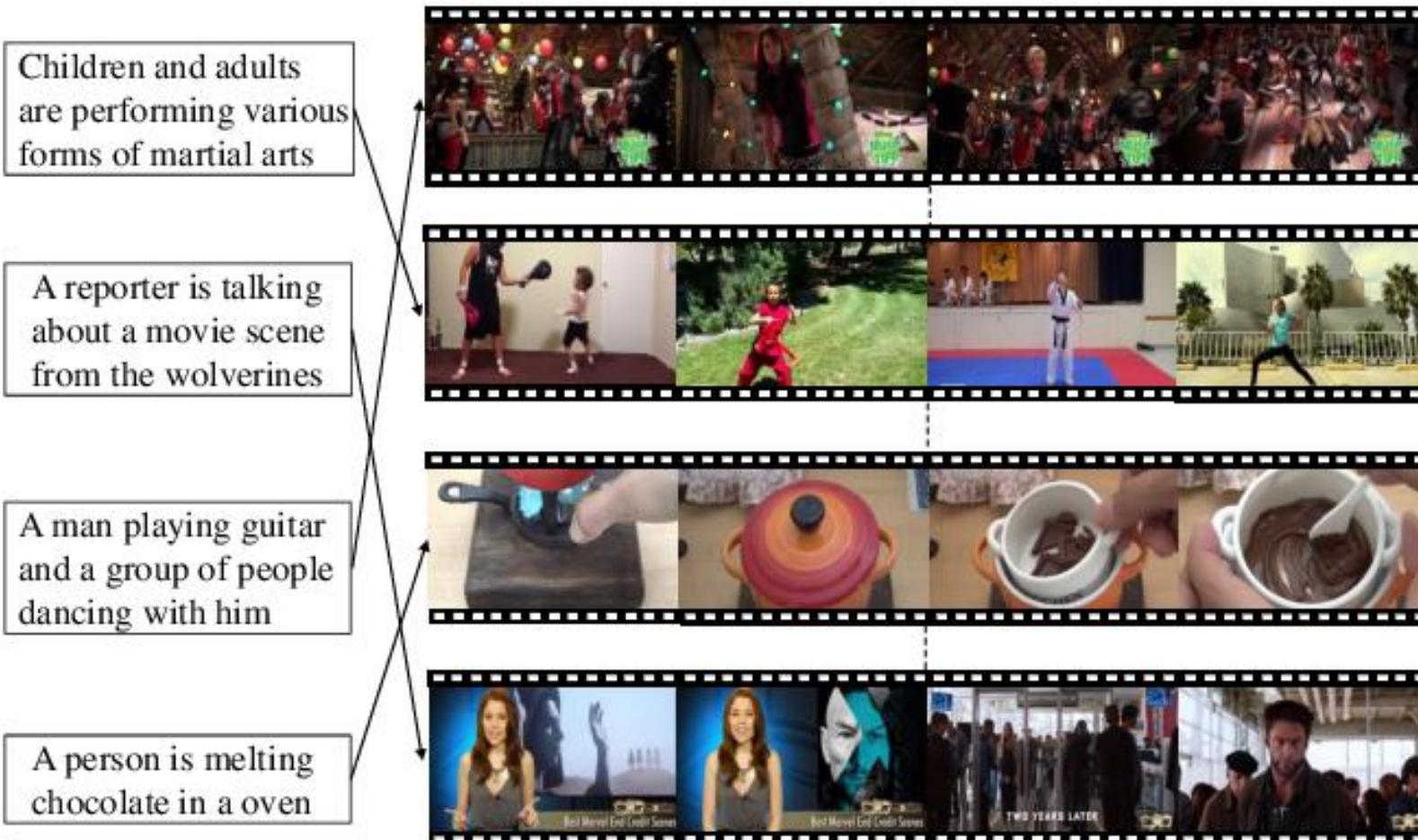
Factor Graph Attention; Schwartz et al.; CVPR'19

Ensemble of MRR and NDCG models for Visual Dialog; Schwartz; (Winner of 2020 VisDial challenge)

Visual Dialog Results

Model	MRR	R@1	R@5	R@10	Mean
LF [10]	0.5807	43.82	74.68	84.07	5.78
HRE [10]	0.5846	44.67	74.50	84.22	5.72
HREA [10]	0.5868	44.82	74.81	84.36	5.66
MN [10]	0.5965	45.55	76.22	85.37	5.46
HieCoAtt-QI [33]	0.5788	43.51	74.49	83.96	5.84
AMEM [44]	0.6160	47.74	78.04	86.84	4.99
HCIAE-NP-ATT [32]	0.6222	48.48	78.75	87.59	4.81
SF-QIH-se-2 [21]	0.6242	48.55	78.96	87.75	4.70
CorefNMN [27]*	0.636	50.24	79.81	88.51	4.53
CoAtt-GAN-w/ $\mathbf{R}_{inte}$ -TF [53]	0.6398	50.29	80.71	88.81	4.47
CorefNMN (ResNet-152) [27]*	0.641	50.92	80.18	88.81	4.45
FGA (VGG)	0.6525	51.43	82.08	89.56	4.35
FGA (F-RCNNx101)	0.6712	54.02	83.21	90.47	4.08
Ensemble of 9 FGA (VGG)	0.6892	55.16	86.26	92.95	3.39

Video retrieval



Baseline: Joint embedding space

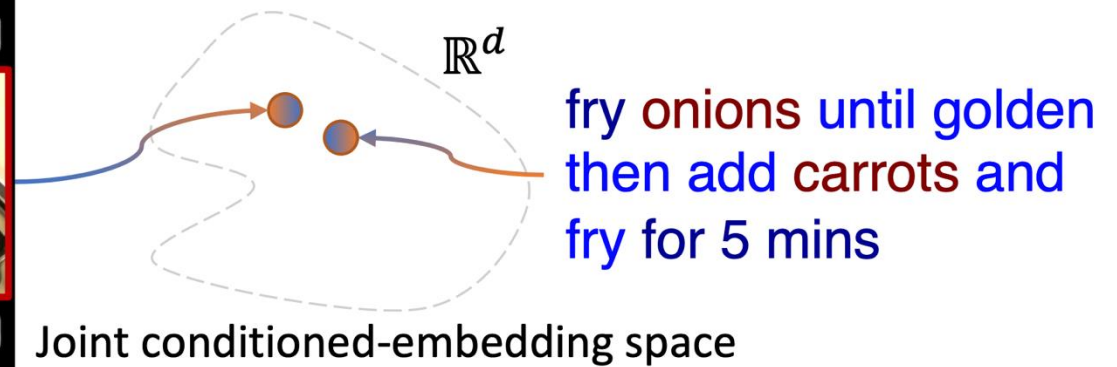
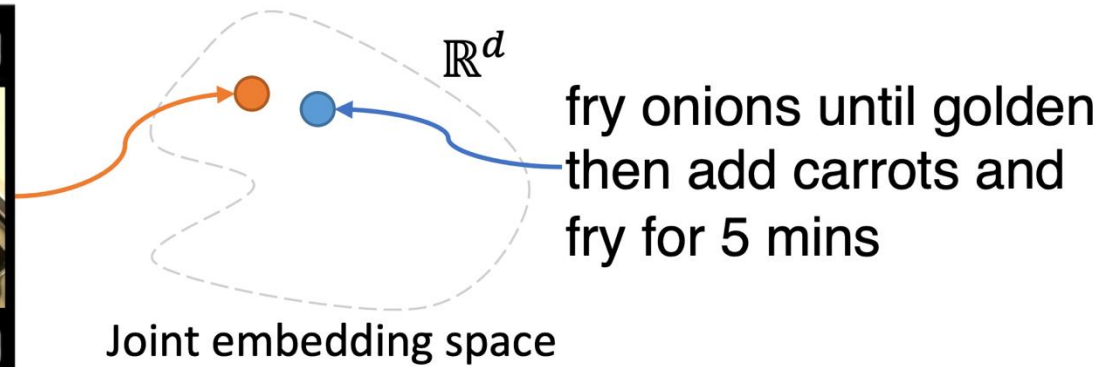


Children and adults
are performing various
types of martial arts

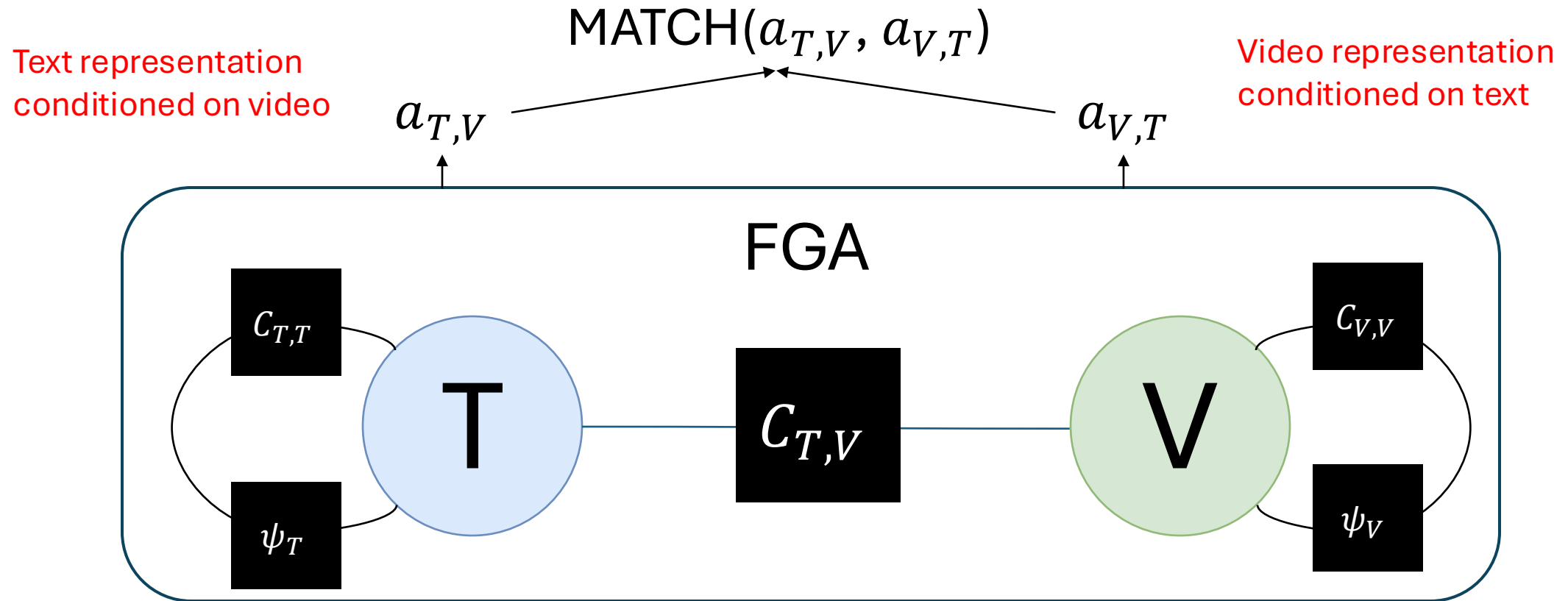


A person is melting
Chocolate in a oven

Conditioned embedding space



Conditioned matching via Attention



A person is melting
Chocolate in a oven



The attention picks relevant-to-query frames and words

Low frame-attention belief



fry onions until golden then add carrots and fry for 5 mins

Low frame-attention belief



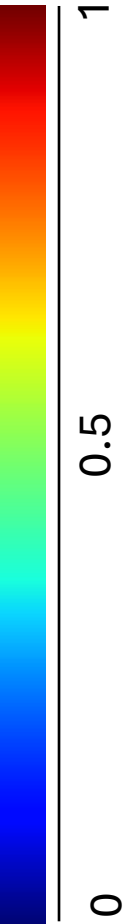
High frame-attention belief



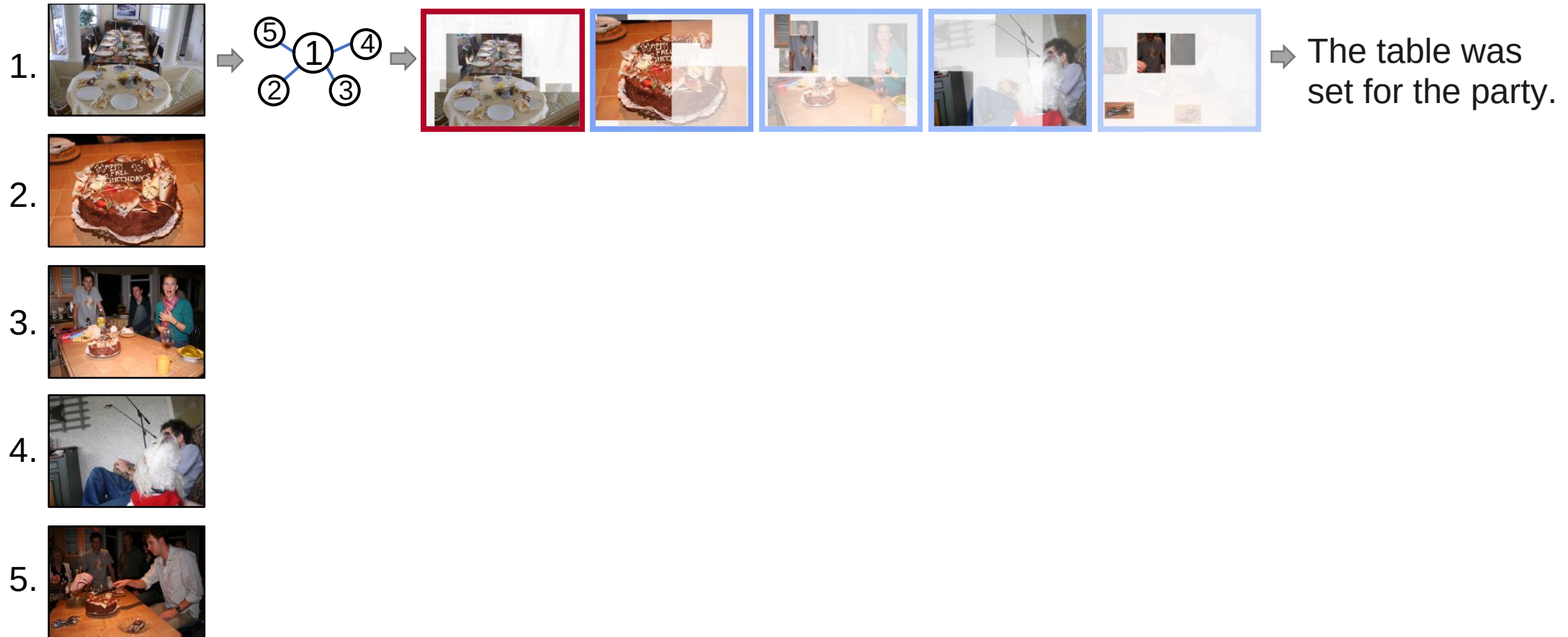
High frame-attention belief



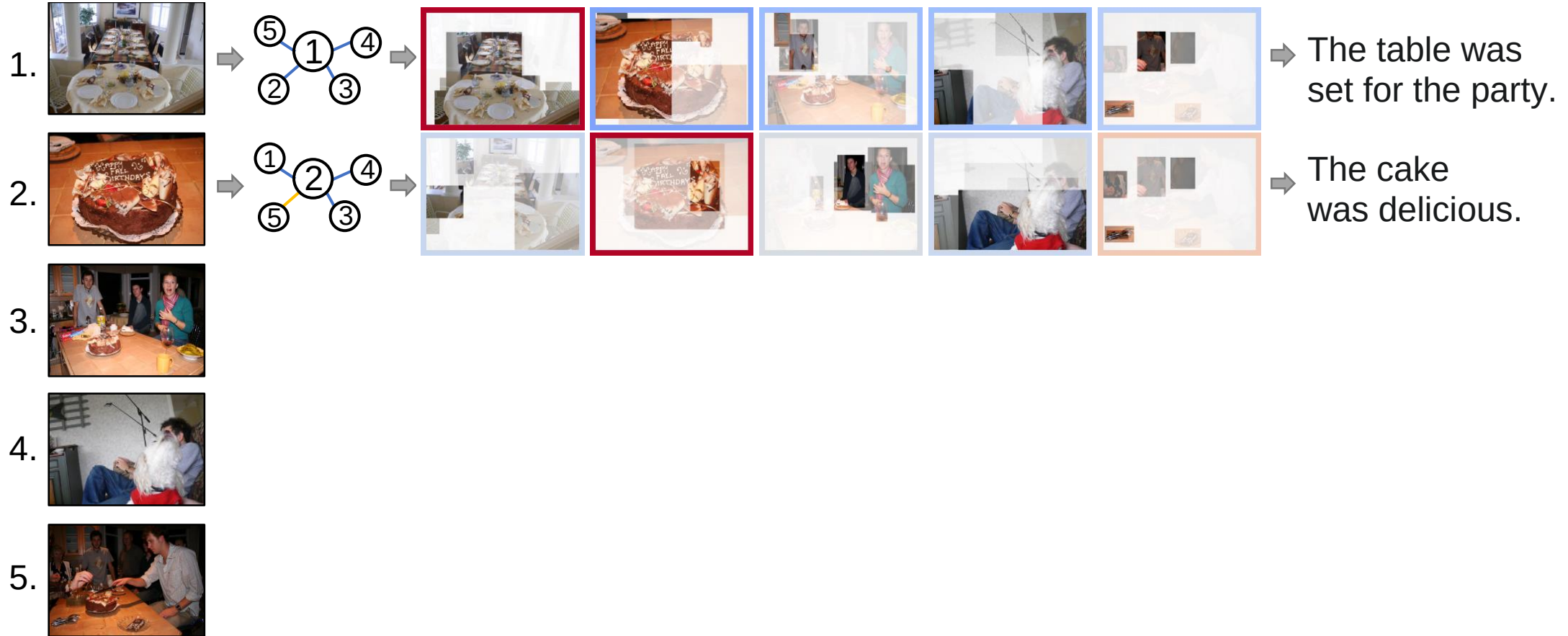
Two women are standing outside the bathtub spraying the dog with a hose.



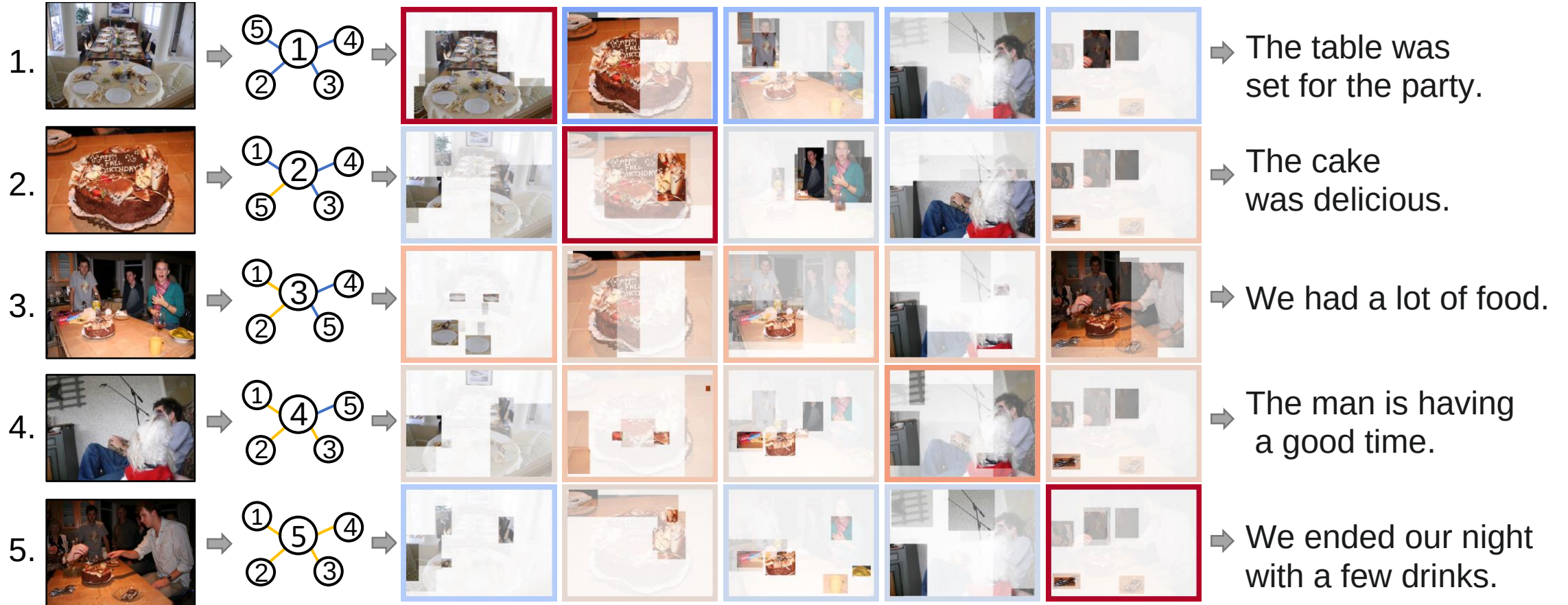
Attention for Visual Storytelling



Attention for Visual Storytelling

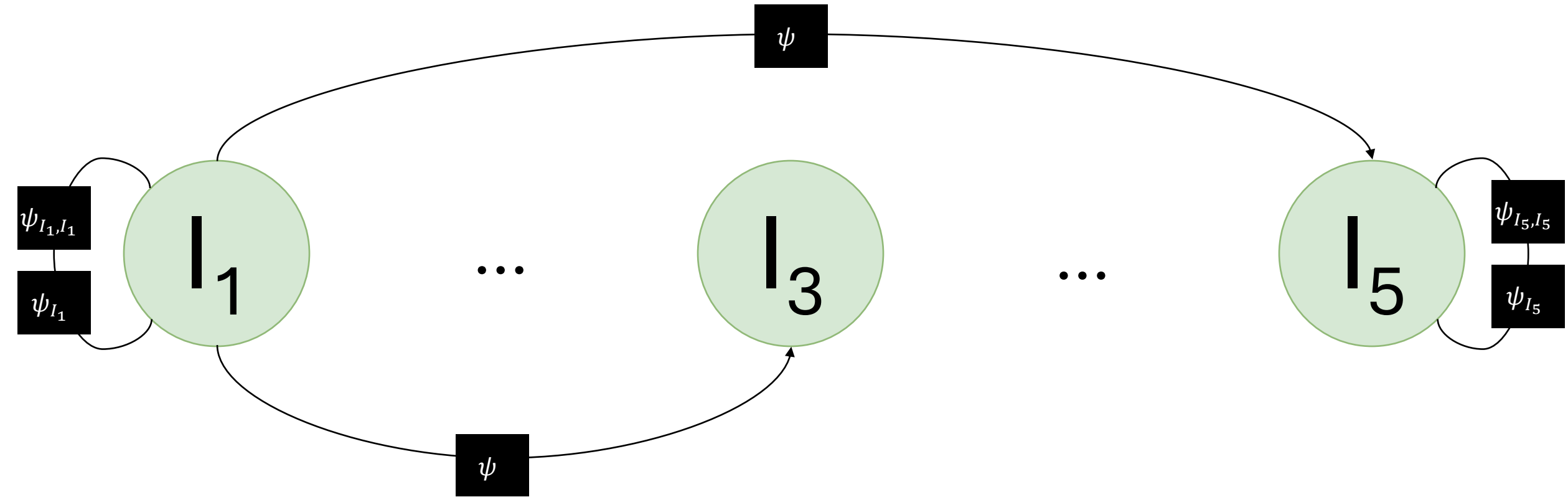


Attention for Visual Storytelling



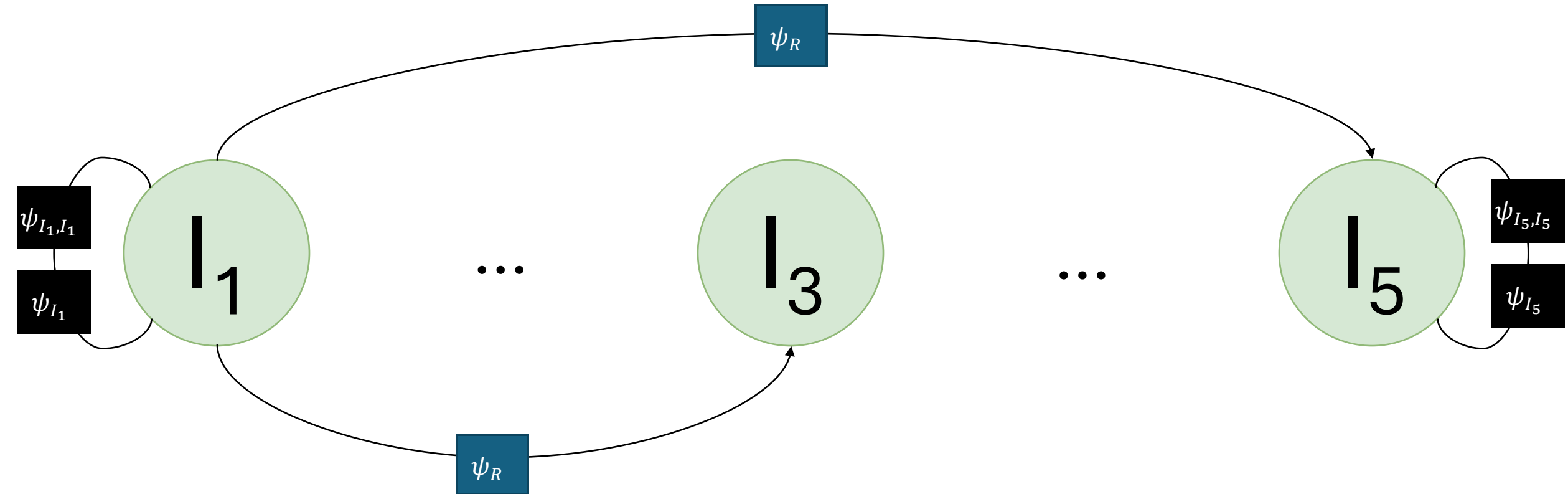
Sharing-weights

- Use the same factor for image interactions



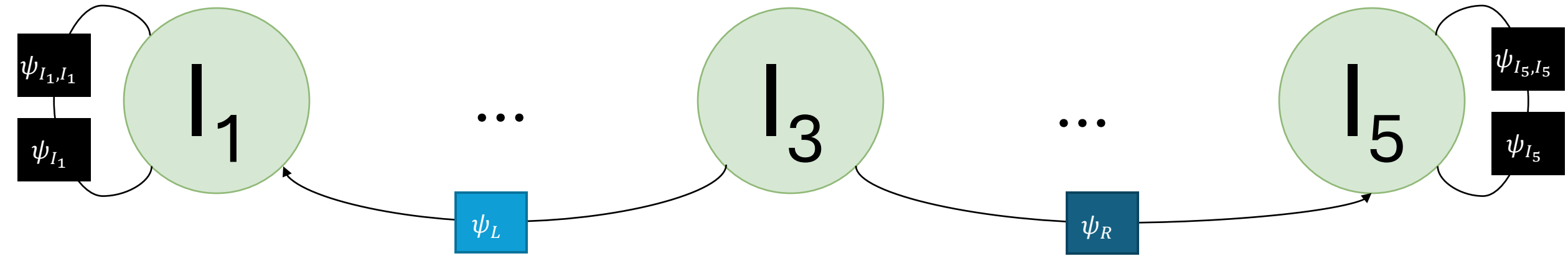
Order-aware factors

- 1st sentence



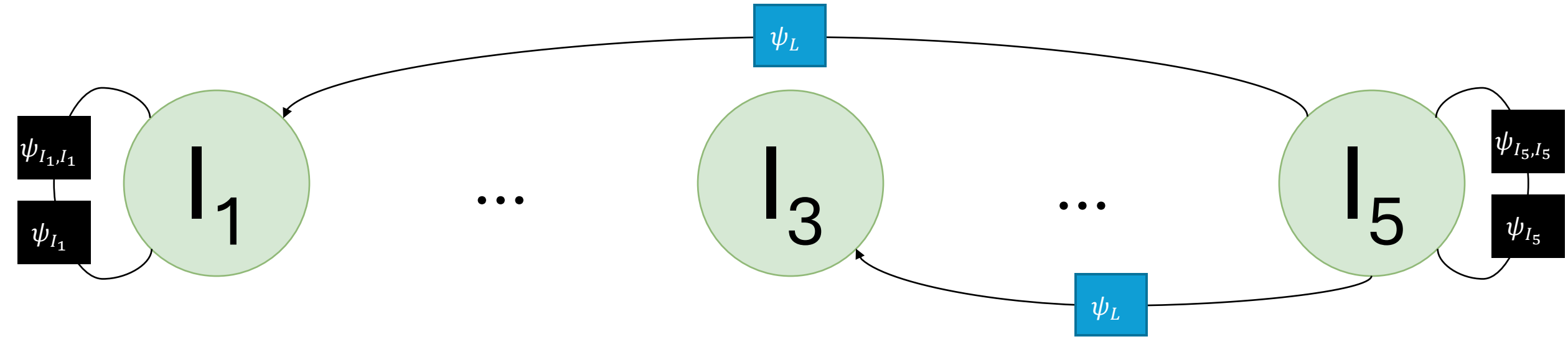
Order-aware factors

- 3rd sentence

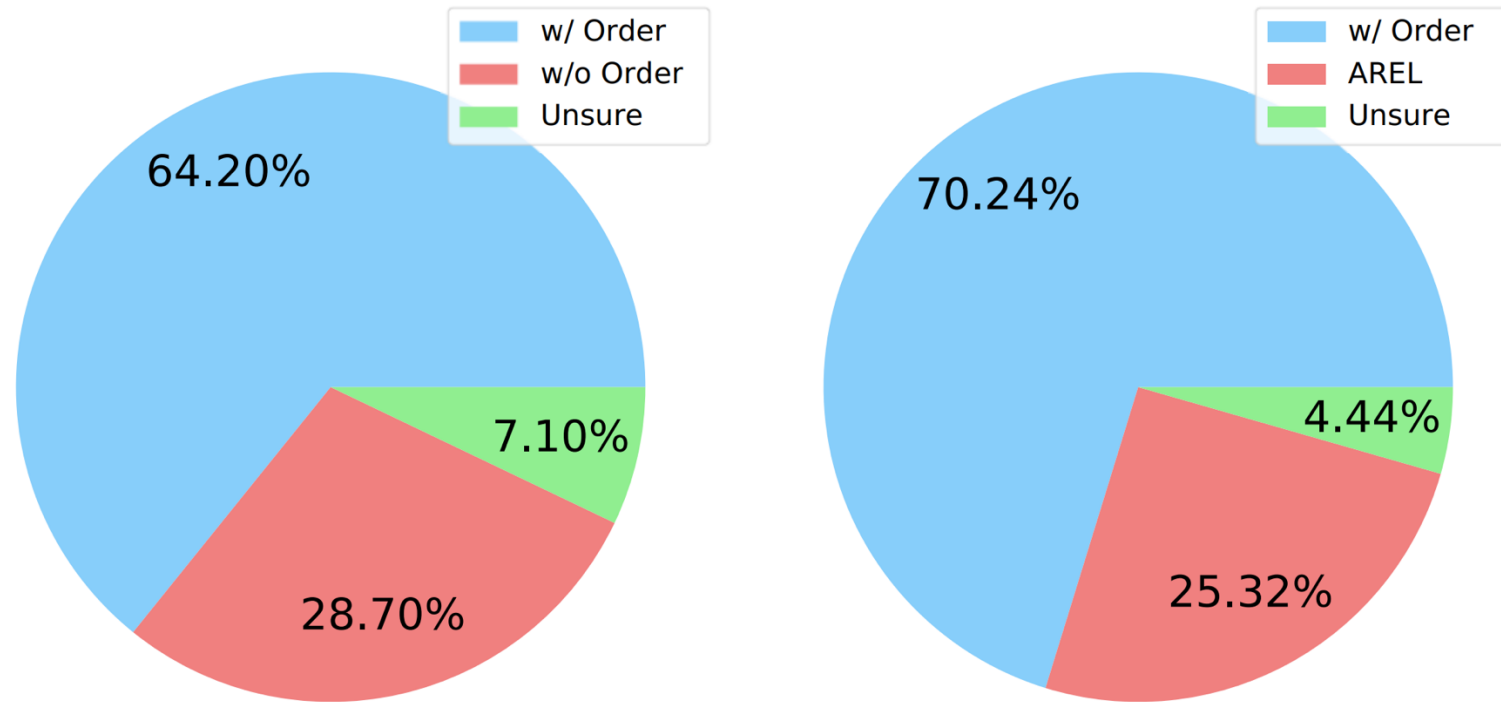


Order-aware factors

- 5th sentence



Order-aware helps coherence



(b) Coherence property comparison.

Visual Navigation*

- Looking for a Toaster



Step 0



Step 1



Visual Navigation*

- Looking for a Toaster

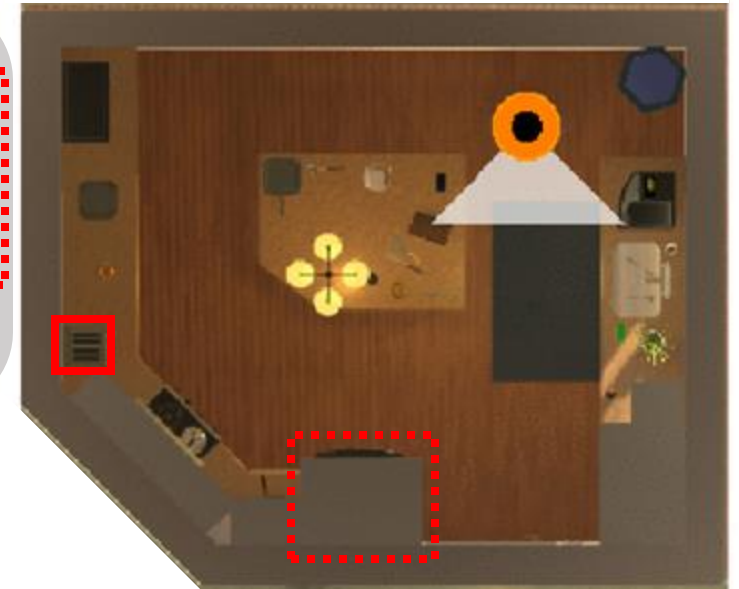


Looking for a Toaster

- Focus on the **fridge**



Step 16



Looking for a Toaster

- Focus on the fridge
- Focus on the **toaster**



Step 22



Questions